AKADEMIKA

BEL-LIT

LINEAR IC TRAINER

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

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It is our vision to provide you with the product you need for training ensuring lasting reliability & quality

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- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

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As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr.No.15/8/1,Unit No. 9 Kruti Industrial Estate,

Opp Karishma Society, Off Karve Road, Kothrud,

Pune – 411038, India

Contact No.: +91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.

- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.

- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.

- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.

- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

TECHNICAL SPECIFICATIONS

- **On board Linear ICs**

  - High Speed Comparator IC710.
  - Function Generator IC ICL8038.
  - Phase Locked Loop IC NE 565.
  - Fixed Voltage regulator LM7805, LM7905.
  - Variable Voltage Regulator IC317, IC337 and LM723.
  - Opto Coupler MCT2E

- **On Board Interfaces**

  - Resistor Bank
  - Capacitor Bank
  - Zener Diode
  - IC Socket
  - Potentiometers
  - 20 pin ZIF socket

- **Power Supply**

  Fixed Power Supply ±25V, ±12V, GND

- **Bread Board**

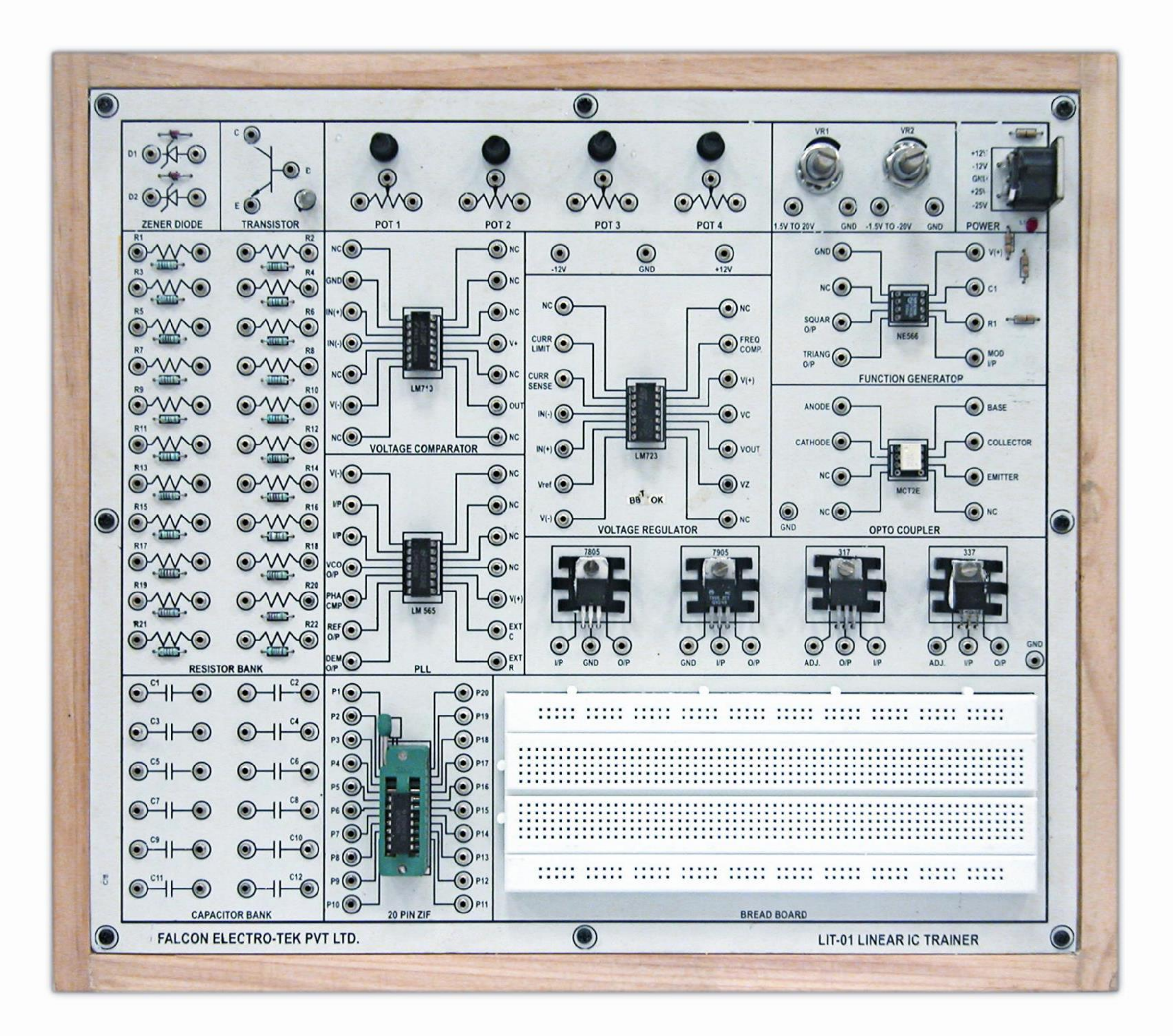
  All Interconnections are made using 2mm patch cords.

Note: Reading Tables and Graphs provided in this manual are taken on sample trainer board hence actual readings and graphs may differ from the readings and graphs shown in this manual. Readings and graphs in this manual are to be treated as sample only for reference purpose to get idea of characteristics of the respective component.

Deliverables

74LS90	: 01 No.
Short Links	: 15 No's
Long Links	: 05 No's
Single Stand Wire	: 02 mtr
BEL-LIT EXPT	: 01 No.

FUNCTIONAL BLOCK



INTRODUCTION

The Linear IC Trainer is designed to cover wide range of analog applications using monolithic ICs such as Op-Amp (LM710), PLL (NE565), opto-coupler (MCT2E) and regulators (LM723, LM7805, LM7905, LM317, LM337). The main idea is to introduce to the students the functions and the usage of the linear ICs having pre defined functions. Once the operation and design methodology is known, the students will be able to build applications based on these fundamental circuits.

To achieve the flexibility in design, various resources such as resistor, capacitor, diode and Potentiometer banks of different values are provided on board. Bread board area allows construction of circuits using external components along with on-board resources.

This trainer system is provided with descriptive experimental manual.

EXPERIMENT NO.1

EXPERIMENT NO. 1

NAME:-

To Study Parameters of High Speed Comparator LM710

OBJECTIVE:-

To study the parameters such as input offset voltage, input bias & offset current, response time, slew rate and CMRR.

EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- DMM
- Function generator

THEORY:-

1) Input Offset Voltage

In an operational amplifier, the input offset voltage is the difference in the voltage between the + and - input pins when negative feedback is applied. In a comparator, the input offset voltage is the comparison voltage (threshold voltage) error. Although ideally 0, in an operational amplifier, there is a difference between V_{BE} of the + and - input transistors in the input-stage differential amplifier, which becomes the input offset voltage.

In actual use; for example when using an operational amplifier with an input offset voltage of $V_{IO} \leq \pm 5mV$, if ×100 amplification is created, there will be an error of up to ±5×100= ±500mV at the output. Therefore, when using a circuit that handles very small input signals, you must either select a product with a small input offset voltage, or adjust the offset externally.

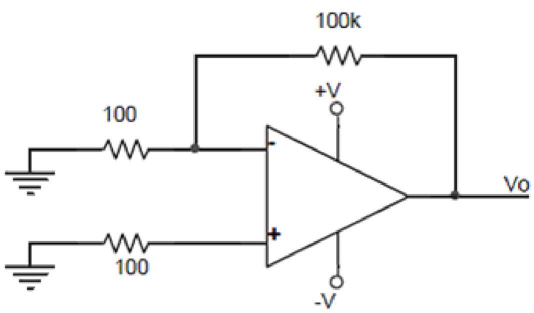


FIG.1 Input offset voltage measurement.

PROCEDURE:-

- Connect the circuit as shown in Figure 1.
- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7v using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Measure the DC output voltage at pin 9 using multimeter and record the result in Table Calculate the input offset voltage using the formula Vi = Vout / 1000 and record the value in table 2

V_{out}	$V_{in} = V_{ou}t/1000$

Table 2

Example

Voltage at pin 11 = 2.1V

Therefore input offset voltage by above formula =

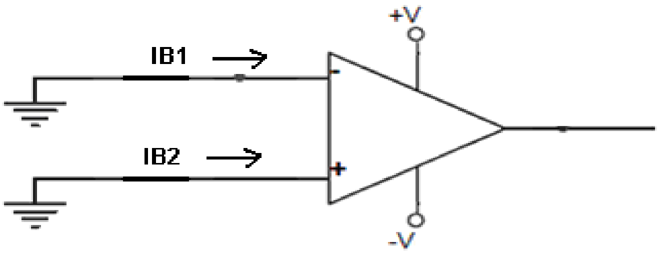
Vi = Vout / 1000

= 2.1V / 1000

= 2.1 mV

2) a) Input Offset Current

It is defined as the algebraic difference between the currents into the inverting and non inverting terminals is refereed to as input offset current (see figure below). In the form of an equation $\mathbf{I_{io} = | I_{B1} - I_{B2} |}$



b) Input Bias Current

Input Bias current $\mathbf{I_B}$ is the average of the currents that flows into the inverting and non inverting input terminals of the comparator. In equation form,

$$\mathbf{I_B = | I_{B1} + I_{B2} | / 2}$$

PROCEDURE:-

- Connect the circuit as shown in Figure 3.
- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7v using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.

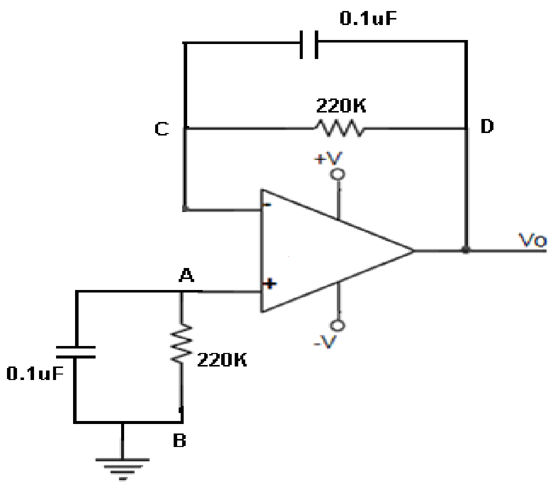
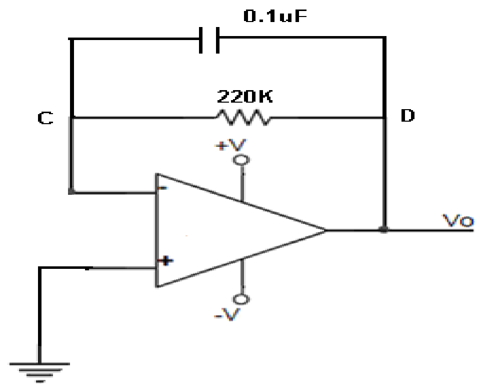
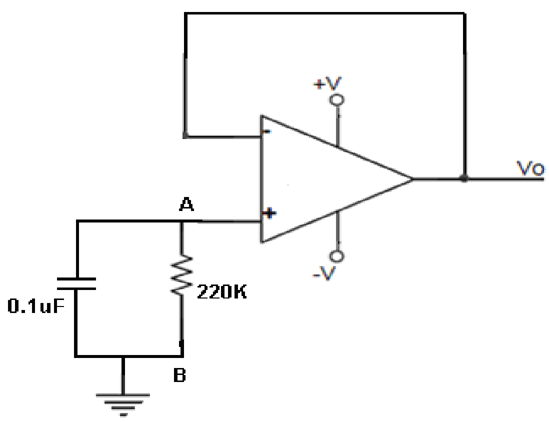


FIG3: Input bias and input offset current

- To measure I_{B1} short point A and B. the equivalent circuit is as shown below.



- Measure output voltage and mark as Vo". Calculate the I_{B1} using formula,
  I_{B1} = Vo"/ 220K.
- To measure $\mathbf{I_{B2}}$ short point C and D. the equivalent circuit is as shown below.



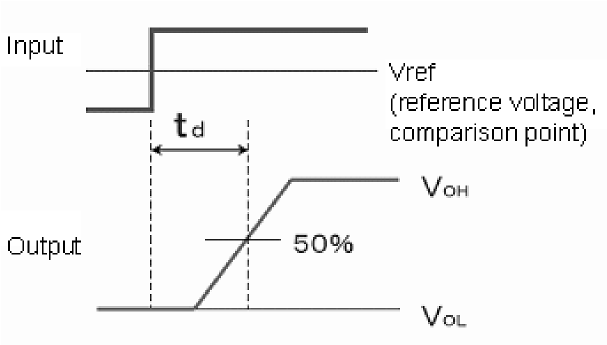
- Measure the output voltage and mark as Vo'. Calculate the I_{B2} using formula,
  I_{B2} = Vo'/ 220K.
- Now by using I_{B1} and I_{B2} calculate input offset current $\mathbf{I_{io} = | I_{B1} - I_{B2} |}$ and input bias current $\mathbf{I_B = | I_{B1} + I_{B2} | / 2}$

EXAMPLE

Vo" is measured as 2.4V. Therefore I_{B1} = 2.4 / 220K
I_{B1} = 10.4uA.
Vo' is measured as -0.5V. Therefore I_{B2} = -0.5 / 220K
I_{B2} = -2.27uA.
Input bias current I_B can be calculated by using I_{B1} & I_{B2}.
Thus Input bias current = |10.4| + | 2.27| / 2
= 6.585uA.
And Input offset current = |10.4| - | 2.27|
= 8.63uA

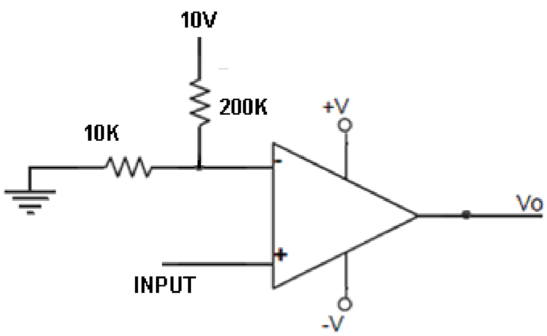
2) Response Time

The response time prescribes the time between when a signal input to the comparator exceeds the reference voltage (V_{ref}) and when the output amplitude reaches 50%. See figure below.



PROCEDURE:-

- Do the connection as shown in fig below.

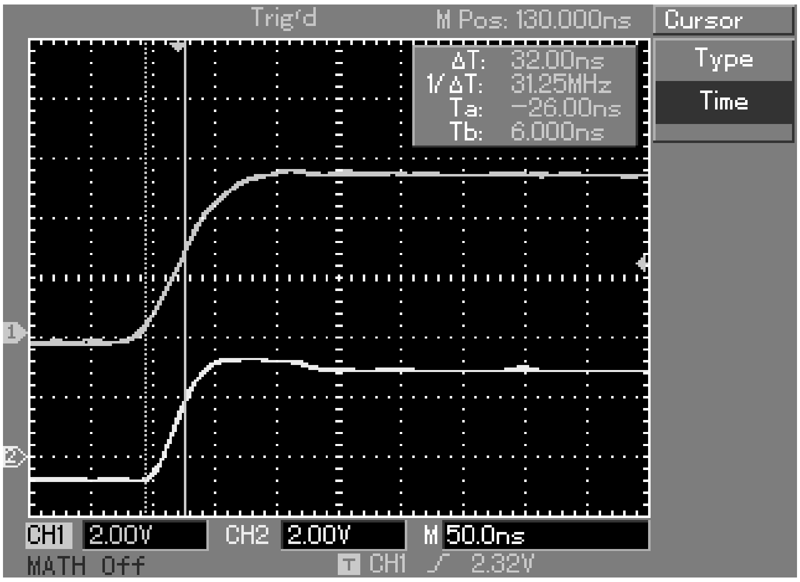


- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 5Vdc square wave of frequency 1 KHz from function generator and connect to pin 3 of LM710.
- Observe the output of comparator on DSO.
- As per definitions set the markers and calculate the response time.

EXAMPLE

Due to voltage divider network at inverting input of comparator reference voltage at that pi is set to 0.5V. Thus set the first marker at 0.5V on input waveform & set second marker at 50% level of output waveform. Refer below fig.

The calculated response time is around 32nS.

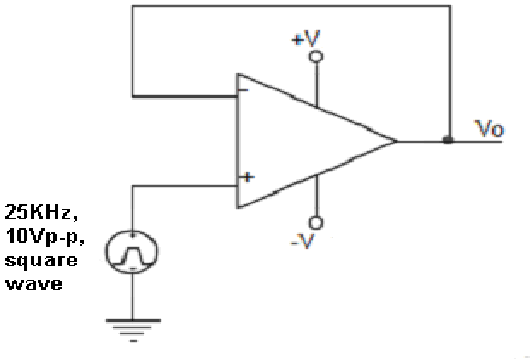


3) Slew Rate

This is the minimum rate of change of voltage per unit time. SR = dv_o / dt_{max}. It is expressed in V/u sec. SR is measured in the unity gain voltage follower circuit. The SR is measured in the amplifier driven by high frequency wave of sufficient magnitude. The SR is slope of the transition between output levels frequently. Positive and negative swings have different SR and both must be examined. In such case lower SR is commonly specified.

PROCEDURE:-

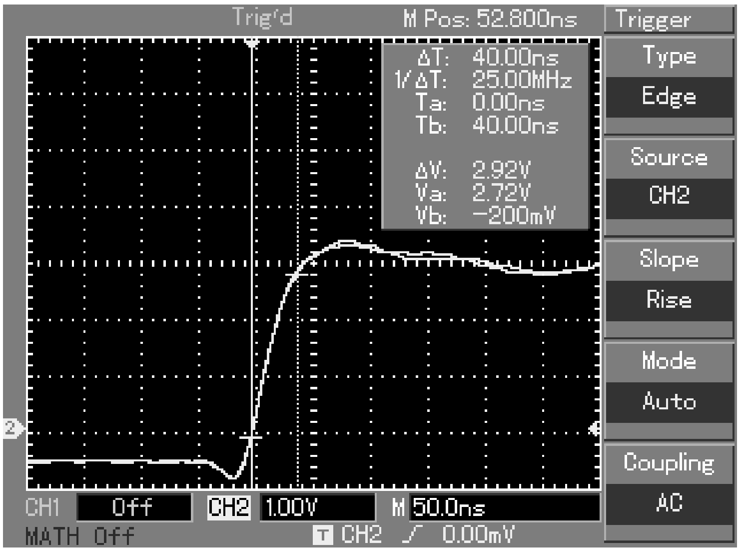
- Do the connection as shown in fig below.



- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 10Vp-p square wave of frequency 25 KHz from function generator and connect to pin 3 of LM710.
- Observe the output of comparator on DSO.
- First measure the voltage swing at rising edge of the signal. Set the markers at 10% and 90% level of signal. (Refer fig in example below). Also measure the voltage swing at falling edge of the signal.
- Express SR in V/usec.

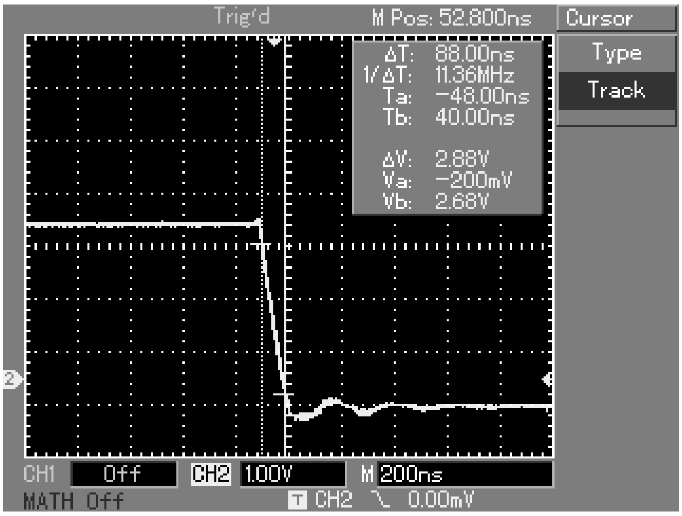
EXAMPLE

Fig below shows the voltage swing at rising edge of the output signal.



SR for rising or positive swing is 2.92V / 40nS.

Fig below shoes the SR for falling or negative edge of the signal.



SR for rising or negetive swing is 2.88V / 88nS.

4) Common Mode Rejection Ratio (CMRR)

It is defined as the ability of a differential amplifier to reject a common-mode signal is expressed by a ration called **Common Mode Rejection Ratio**, denoted as **CMRR**".

CMRR is defined as the ratio of the differential voltage gain Ad to common mode voltage gain Ac.

CMRR = A_d / A_c

Ideally A_c is zero. Hence, the ideal value of CMRR is ∞.

Theoretical Background of CMRR

The common mode rejection ratio (CMRR) is defined as $CMRR = \frac{A_d}{A_c}$ where A_d and A_c are the difference mode and common mode gains respectively. The voltage between the inverting terminal and non-inverting terminal is

$$e = V_1 - V_2$$

From superposition, the voltage is

$$V_1 = V_S \frac{R_2}{R_1 + R_2} + V_O \frac{R_1}{R_1 + R_2}$$

and it can be shown that

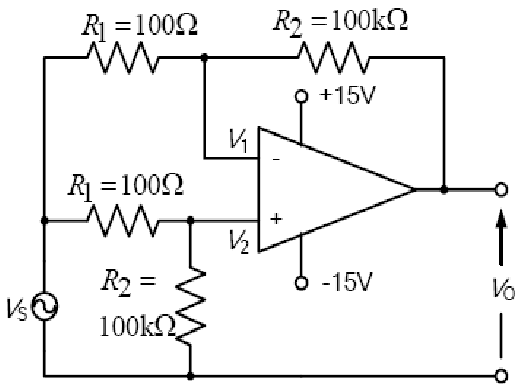


FIG. 4 CMRR

$$V_2 = V_S \frac{R_2}{R_1 + R_2}$$

Then we have

$$e = V_O \frac{R_1}{R_1 + R_2}$$

Let V_C be the common-mode voltage, then

$V_C = V_2 + \frac{e}{2} = V_S + \frac{e}{2}$ when $R_2 >> R_1$. Since $V_O = -A_d e + A_C V_C$, we have

$$V_O = -A_d \frac{V_O R_1}{R_1 + R_2} + A_C (V_S + \frac{V_O R_1}{2(R_1 + R_2)})$$

When A_d **>>** A_c and $A_d \frac{R_1}{R_1 + R_2} >> 1$, the above equation yields the following expression for the CMRR

$$CMRR = \frac{A_d}{A_C} = \frac{V_S}{V_O} \frac{R_1 + R_2}{R_1}$$

For the values of *R*1 and *R*2 used in the circuit shown in the above figure,

$$CMRR = 1001 \frac{V_S}{V_O}$$

PROCEDURE:-

- Do the connection as shown in fig 4.
- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 5Vp-p square wave of frequency 1 KHz from function generator and connect to LM710 as shown in fig 4.
- Measure the output voltage using DMM or DSO. And calculate the CMRR using Equation 1. To get the CMRR in dB use following formula,

- CMRR = 20 log [1001 (V_S/V_O)] dB.
- Repeat the calculations for different frequency.

EXAMPLE

Frequency	V_O	$CMRR = 1001\dfrac{V_S}{V_O}$, V_S =5V	CMRR in dB
10 Hz	118mV	42415	92
100 Hz	118mV	42415	92
1 kHz	110mV	45500	93
10 kHz	95mV	52684	94

EXPERIMENT
NO.2

EXPERIMENT NO. 2

NAME:-

To Study LM710 as Comparator

OBJECTIVE:-

To study the LM710 as voltage comparator

EQUIPMENTS:-

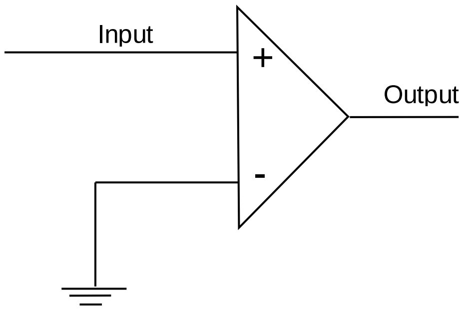
- BEL-LIT board
- Power supply
- Patch chords
- DSO
- Function generator

THEORY:-

The LM710 series are high-speed voltage comparators intended for use as an accurate low-level digital level sensor or as a replacement for operational amplifiers in comparator applications where speed is of prime importance. The circuit has a differential input and a single-ended output with saturated output levels compatible with practically all types of integrated logic The device is built on a single silicon chip which insures low offset and thermal drift The use of a minimum number of stages along with minority-carrier lifetime control (gold doping) makes the circuit much faster than operational amplifiers in saturating comparator applications In fact the low stray and wiring capacitances that can be realized with monolithic construction make the device difficult to duplicate with discrete components operating at equivalent power levels. The LM710 series are useful as pulse height discriminators voltage comparators in high-speed AD converters or go no-go detectors in automatic test equipment. They also have applications in digital systems as an adjustable-threshold line receiver or an interface between logic types In addition the low cost of the units suggests them for applications replacing relatively simple discrete component circuitry Schematic and Connection Diagrams

PROCEDURE:-

- Do the connection as shown in fig below.

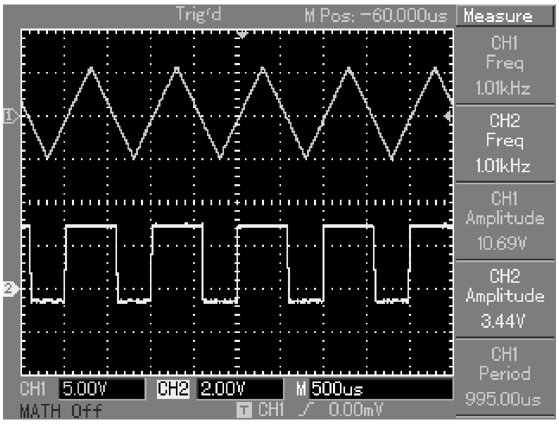


- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 10Vp-p triangular wave of frequency 1 KHz from function generator and connect to pin 3 of LM710.
- Observe the output on DSO at pin 9 of LM710.
- O/p of the comparator switches to Positive level when i/p is greater than 0v and o/p of the comparator switches to Negative level when i/p is less than 0v

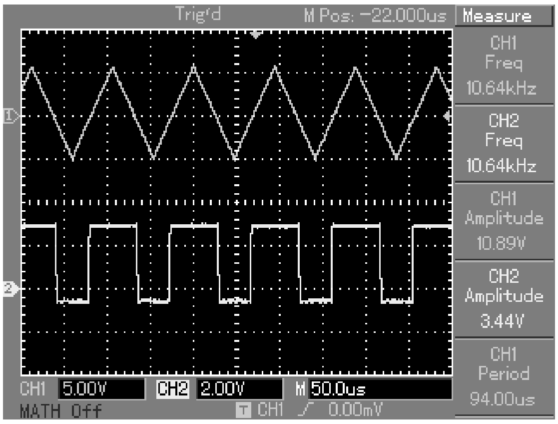
OBSERVATION & WAVEFORMS:-

Output of comparator is up to 3.2V when it is switches to positive level and -0.5V when it switches to negative level.

1) Comparator output for input frequency of 1 KHz

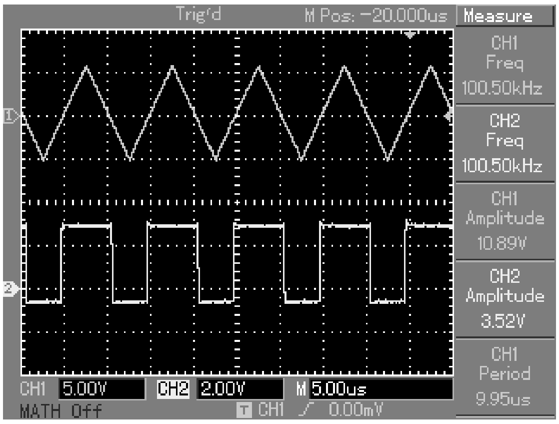


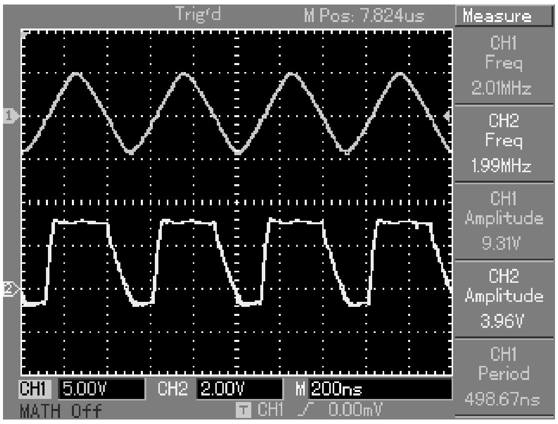
2) Comparator output for input frequency of 10 KHz



3) Comparator output for input frequency of 10 KHz.

4) Comparator output for input frequency of 2MHz.


Trig'd
M Pos: -20.000us
Measure
CH1
Freq
100.50kHz
CH2
Freq
100.50kHz
CH1
Amplitude
10.89V
CH2
Amplitude
3.52V
CH1
Period
9.95us
CH1 5.00V CH2 2.00V M 5.00us
MATH Off
CH1 0.00mV


Trig'd
M Pos: 7.824us
Measure
CH1
Freq
2.01MHz
CH2
Freq
1.99MHz
CH1
Amplitude
9.31V
CH2
Amplitude
3.96V
CH1
Period
498.67ns
CH1 5.00V CH2 2.00V M 200ns
MATH Off
CH1 0.00mV

EXPERIMENT NO.3

EXPERIMENT NO. 3

NAME:-

To Study LM710 as Level Detector

OBJECTIVE:-

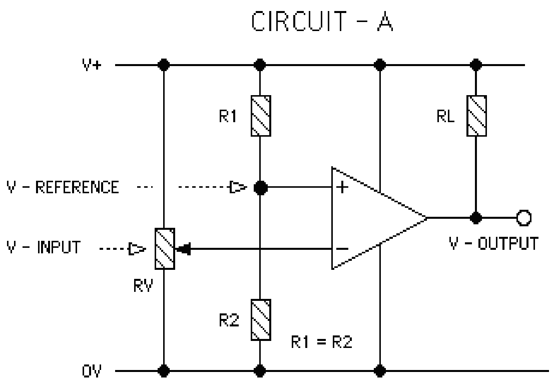
To study the LM710 as Level Detector

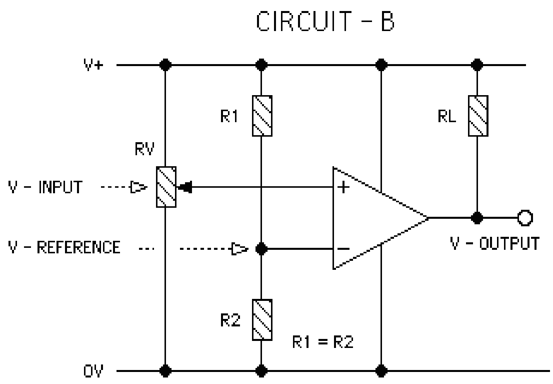
EQUIPMENTS:-

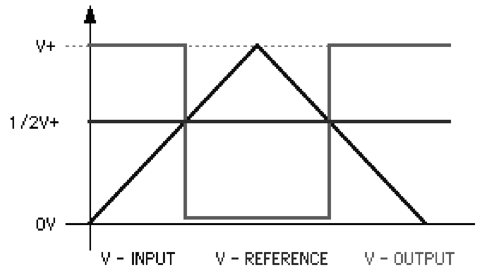
- BEL-LIT board
- Power supply
- Patch chords
- DSO
- DMM
- Function generator

THEORY:-

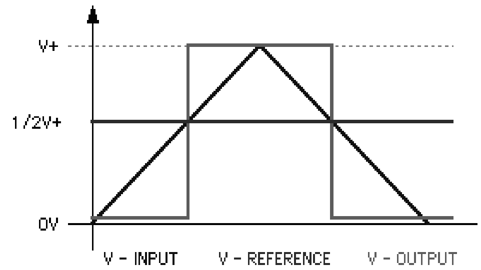
The following figure shows the two simplest configurations for voltage level detectors. The diagrams below the circuits give the output results in a graphical form.
For these circuits the REFERENCE voltage is fixed at one-half of the supply voltage while the INPUT voltage is variable from zero to the supply voltage.
In theory the REFERENCE and INPUT voltages can be anywhere between zero and the supply voltage but there are practical limitations on the actual range depending on the particular device used







Output for Circuit- A



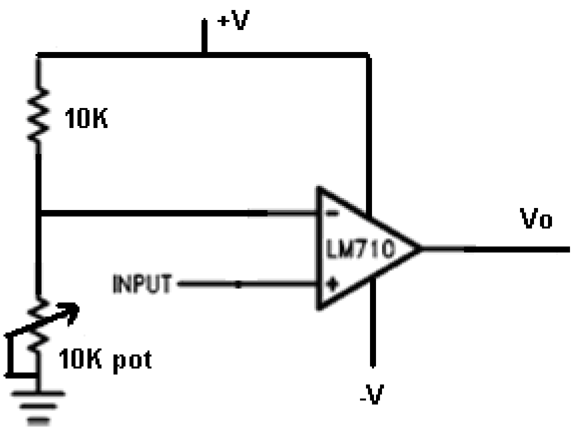
Output For Circuit- B

In Circuit A reference voltage is set at non inverting input & input signal is fed to inverting input of comparator. The output for such configuration is shown on fig above. The output of comparator is inverted.

In Circuit B input is given to non inverting input and reference voltage is connected to inverting input of comparator. The output for this circuit is non inverted form as shown above.

PROCEDURE:-

- Do the connection as shown in fig below.

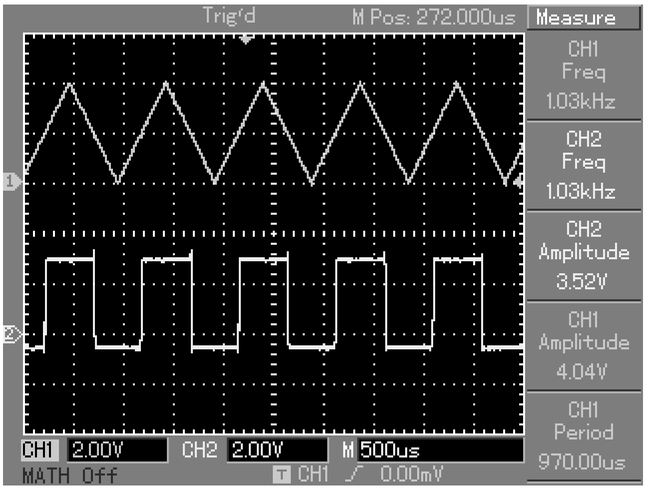


- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 4VDC triangular wave of frequency 1 KHz from function generator and connect to pin 3 of LM710.
- Connect voltage divider network as shown in fig above and set 2V dc as reference voltage by adjusting Pot at inverting input of comparator.

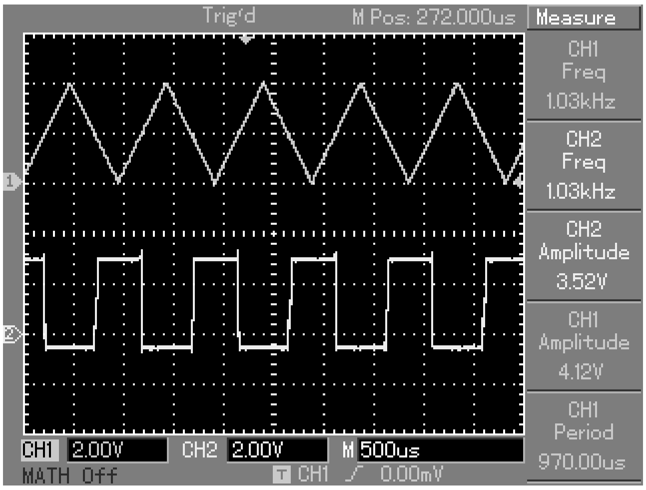
- Observe the output on DSO at pin 9 of LM710.
- Now vary either amplitude of the input signal or reference voltage and observe the effect on output of comparator. Also vary the frequency of the input signal and observe the effect on output.
- To operate comparator in inverting configuration connect input signal to inverting pin (pin 4) and set reference voltage at non inverting pin (pin 2). And repeat the steps from 4 to 6.

OBSERVATIONS AND WAVEFORMS:-

- Output of comparator when operated in non inverting mode.



- Output of comparator when operated in inverting mode.



EXPERIMENT NO.4

EXPERIMENT NO.4

NAME:-

To Study LM710 as Schmitt Trigger

OBJECTIVE:-

To study the application of LM710 as voltage comparator

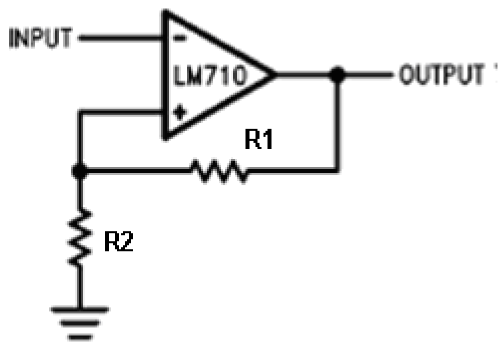
EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- DMM
- Function generator

THEORY:-

The Schmitt trigger is a comparator application which switches the output negative when the input passes upward through a positive reference voltage. It then uses negative feedback to prevent switching back to the other state until the input passes through a lower threshold voltage, thus stabilizing the switching against rapid triggering by noise as it passes the trigger point.

Schmitt Trigger



In other words it is a circuit which converts irregular shaped waveforms to a square wave pulse. The input voltage triggers the output voltage every time it exceeds certain voltage levels called Upper Threshold Voltage (**V_{UT}**) and Lower Threshold Voltage (**V_{LT}**). These voltages are obtained by using voltage divider R1 and R2. Where voltage across R1 is fed back. Reference threshold voltage across R1 is a variable ref. threshold voltage that depends on value and polarity of output voltage Vo.

When Vo is at positive level, the voltage across R1 is called Upper threshold voltage (**V_{UT}**). The input voltage must be slightly more positive than **V_{UT}** to switch from positive level to negative level.

$$V_{UT} = (R1 / R1+R2) \text{ X } +V$$

When Vo is at negative level, the voltage across R1 is called Lower threshold voltage (**V_{LT}**). The input voltage must be slightly more negative than **V_{LT}** to switch from negative level to positive level.

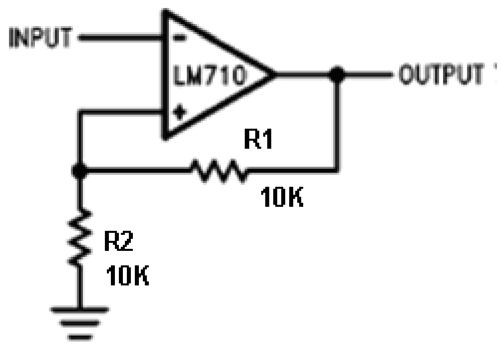
$$V_{LT} = (R1 / R1+R2) \text{ X } -V$$

Moreover, **V_{UT} & V_{LT}** If are made longer than the input noise voltages, the positive feedback will eliminate the false output transitions. Also due to regenerative action of positive feedback it will make Vo to switch faster between positive and negative level.

PROCEDURE:-

- Do the connection as shown in fig below.

Schmitt Trigger

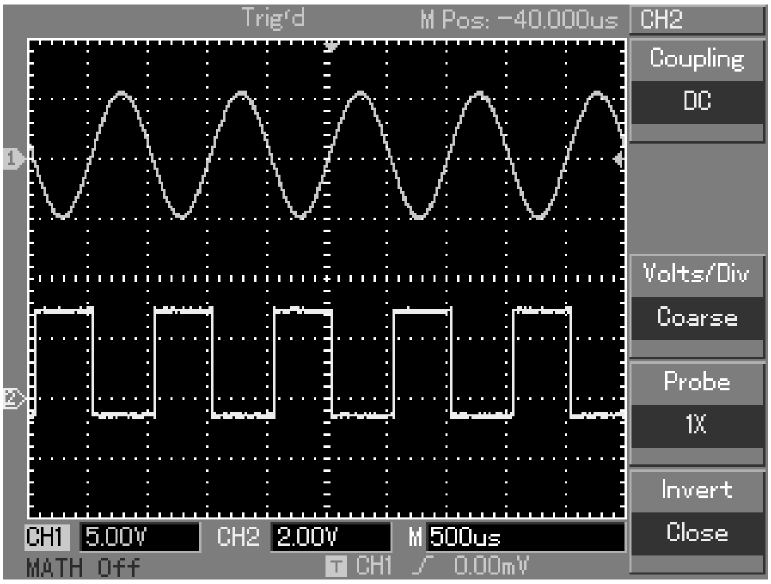


- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Design **V_{UT}** using formula **V_{UT}** = (R1 / R1+R2) X +V. In LM 710 maximum output level is +3.2V, so design **V_{UT}** to 1.6V & assume R1 =10K
- 1.6V = (10K / 10K + R2) X 3.2V

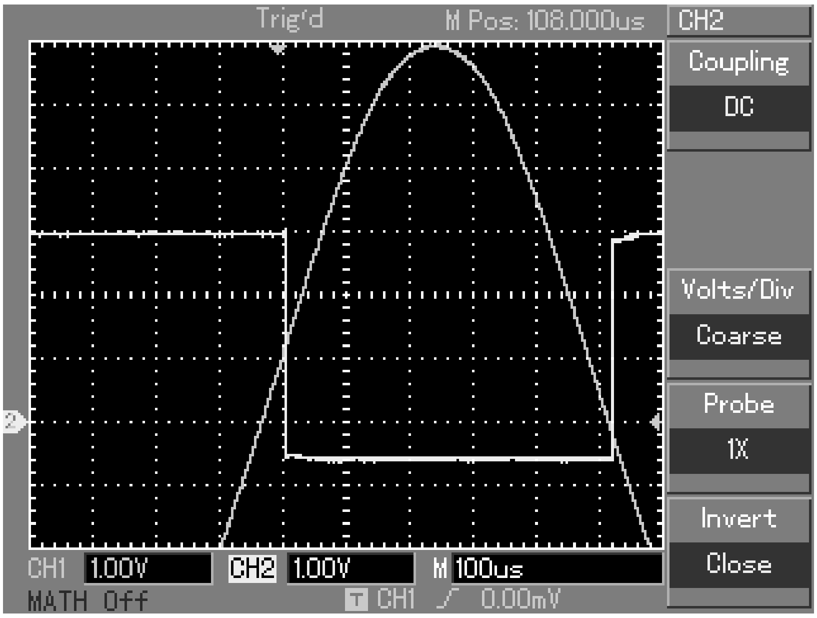
  Thus R2 = 10K
  Similarly calculate **V_{LT}** by considering values of R1 & R2 as 10K.
  V_{LT} = (R1 / R1+R2) X -V.
  In LM710 maximum negative level is -0.5V. Considering -V as -0.5 V calculate **V_{LT}**.
  V_{LT} = (10K / 20K) X -.0.5
  V_{LT} = -0.25V
- Adjust 10Vp-p sine wave of frequency 1 KHz from function generator and connect to pin 3 of LM710.
- Observe the output at pin 11 & compare it with input signal.
- Compare the practical and theoretical **V_{UT}** and **V_{LT}**.

OBSERVATION AND WAVEFORM:-

- Input and output waveforms of Schmitt trigger.



- Measurement of V_{UT} and V_{LT}.



EXPERIMENT NO.5

EXPERIMENT NO. 5

NAME:-

To Study LM710 as Pulse Width Modulator

OBJECTIVE:-

To study the application of LM710 as Pulse Width Modulator

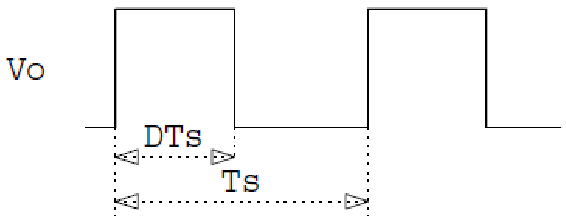
EQUIPMENTS:-

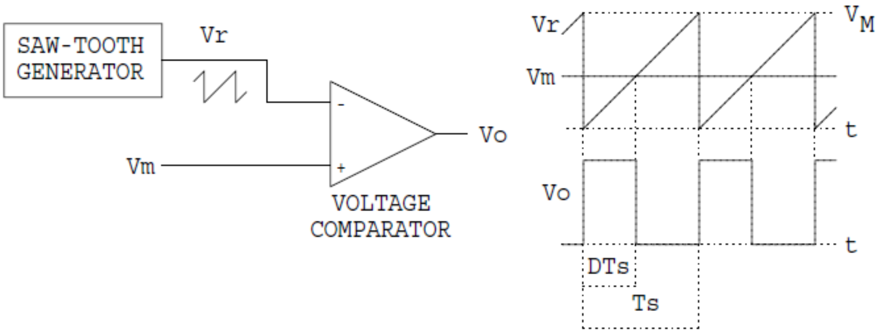
- BEL-LIT board
- Power supply
- Patch chords
- DSO
- DMM
- Function generator

THEORY:-

In the pulse-width modulator to be designed, the output pulse width t_H relative to the period Ts is determined by an analog input voltage Vm,
The pulse width t_H relative to the period Ts is called the duty ratio D of the output waveform. With the analog PWM input Vm, the output duty ratio D is proportional to the to the input voltage
Vm,

$D = t_H / Ts$
$= Vm / VM;$

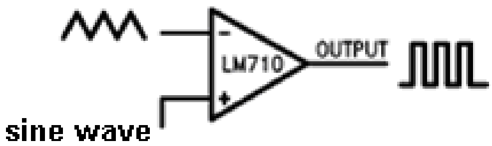




Where VM is a constant. The output duty ratio goes from D = 0 for vm ≤ 0, to D = 1 for vm ≥ VM.

PROCEDURE:-

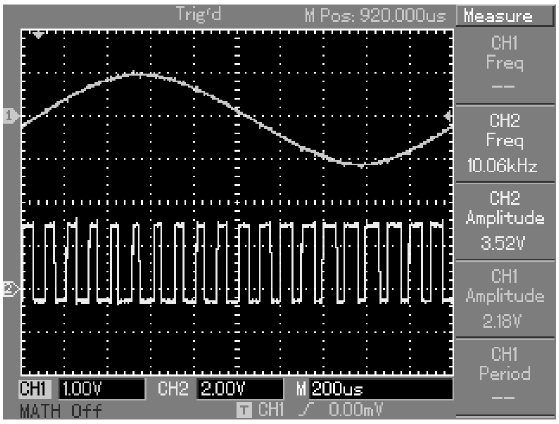
- Do the connection as shown in fig below.



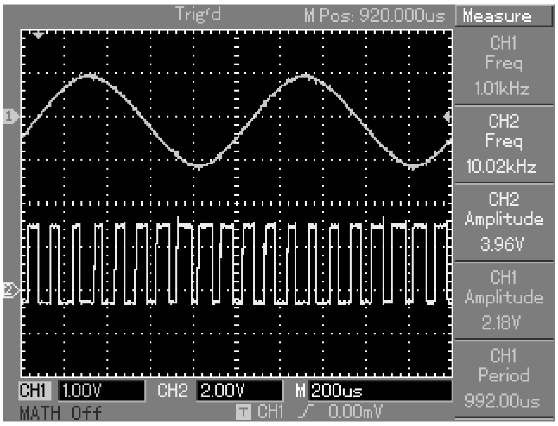
- Adjust +14V using VR1 and connect to pin 11 of LM710 similarly adjust -7V using VR2 and connect to pin 6 of LM710. Connect pin 2 to ground.
- Adjust 6VDC triangular wave of frequency 10 KHz from function generator and connect to pin 4 of LM710. it will act as carrier.
- Adjust 2Vp-p sine wave of frequency 500Hz from function generator and connect to pin 3 of LM710. This will act as modulating signal.
- Observe the output of comparator with respect to input modulating signal.
- Observe the effect of output by varying the frequency and amplitude of the modulating signal.

OBSERVATION AND WAVEFORMS:-

1) PWM output for modulating signal of 500Hz.



2) PWM output for modulating signal of 1 KHz



EXPERIMENT NO.6

EXPERIMENT NO.6

NAME:-

To Study the Function Generator Using ICL8038

OBJECTIVE:-

To design the function generator using ICL8038

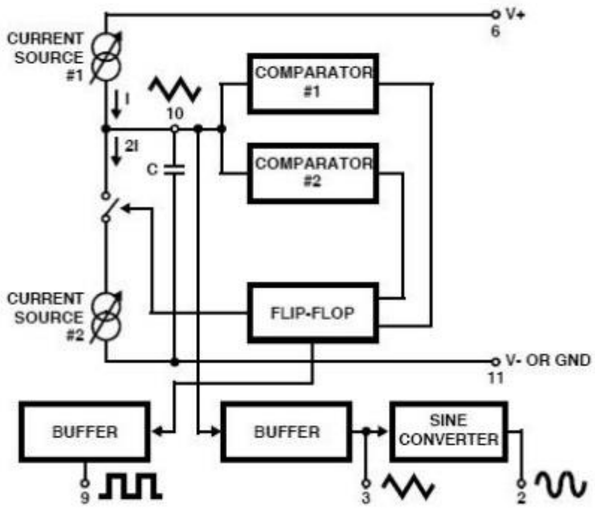
EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- Function generator

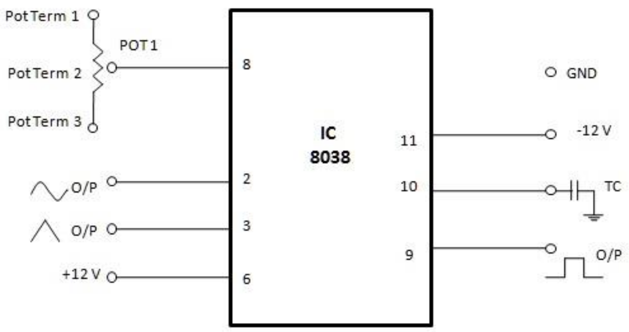
THEORY:-

The ICL8038 Function Generator is a voltage-controlled oscillator of exceptional linearity with buffered square wave and triangle wave outputs. The frequency of oscillation is determined by an external resistor and capacitor and the voltage applied to the control terminal. The oscillator can be programmed over a ten-to-one frequency range by proper selection of an external resistance and modulated over a ten-to-one range by the control voltage, with exceptional linearity.

Internal block diagram of ICL8038

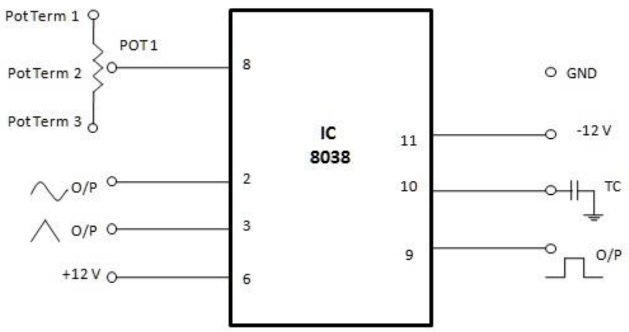


The ICL8038 Function Generator is a general purpose voltage-controlled oscillator designed for highly linear frequency modulation. The circuit provides simultaneous square wave and triangle wave outputs at frequencies up to 1MHz. A typical connection diagram is shown in Figure below.



PROCEDURE:-

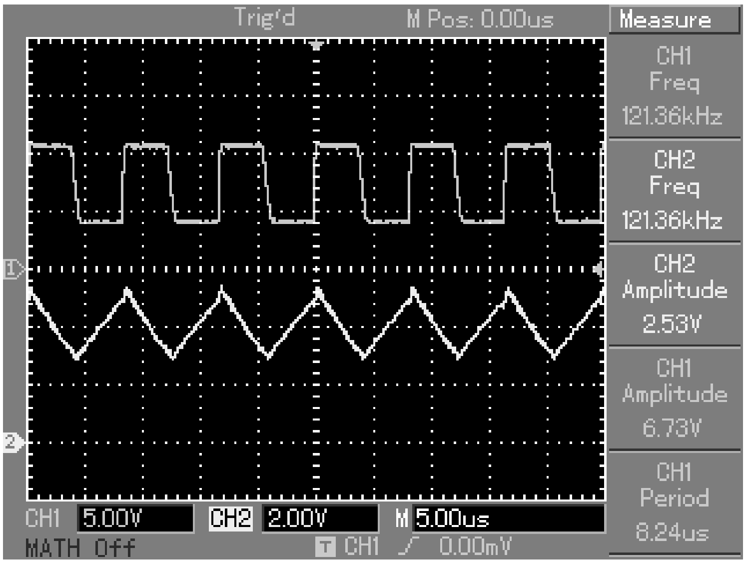
- Do the connection as shown in fig below.



- Connect fixed 12v supply to pin no0 6 of ICL8038 and connect -12V supply to pin no 11.
- Connect POT1 (1K) to pin no 8.
- Connect capacitor C11 (330PF) to pin no 10.
- Observe the output of function generator (Sine, Square & Triangular) on the DSO.

OBSERVATION AND WAVEFORMS:-

The practical value of the frequency of is not same as per theoretical values. It is due to the component tolerances & connecting cables. Also both the outputs are DC shifted from ground.


Trig'd
M Pos: 0.00us
Measure
CH1
Freq
121.36kHz
CH2
Freq
121.36kHz
CH2
Amplitude
2.53V
CH1
Amplitude
6.73V
CH1
Period
8.24us
CH1 5.00V
CH2 2.00V
M 5.00us
MATH Off
T CH1 0.00mV

EXPERIMENT NO.7

EXPERIMENT NO. 7

NAME:-

To Study ICL8038 as Frequency Modulator.

OBJECTIVE:-

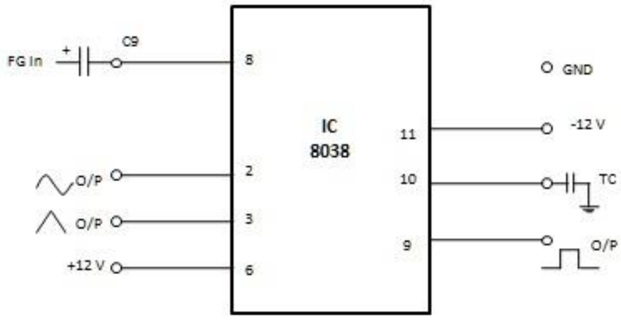
To study the application of ICL8038 as frequency modulator

EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- Function generator

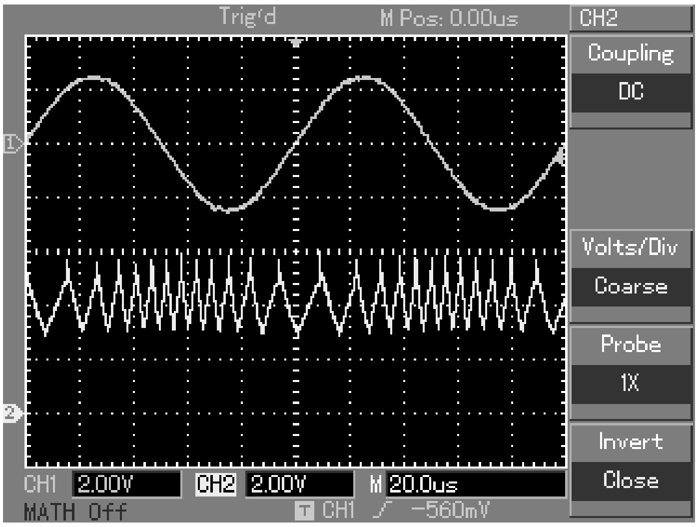
PROCEDURE:-

- Do the connection as shown in circuit below same circuit is used as in EXPT 6.



- Modulating signal of 5 Vp-p 10 KHz sine wave is connect to pin 8 through a 1uF capacitor.
- Observe the frequency modulated output at pin 2 & 3 of ICL8038. Compare the output with modulating signal simultaneously.
- Observe the effect on output by varying the frequency of the modulating signal.

OBSERVATION AND WAVEFORMS:-



EXPERIMENT NO.8

EXPERIMENT NO.8

NAME:-

To Study the Opto-coupler MCT2E.

OBJECTIVE:-

To study and design the application of MCT2E

EQUIPMENTS:-

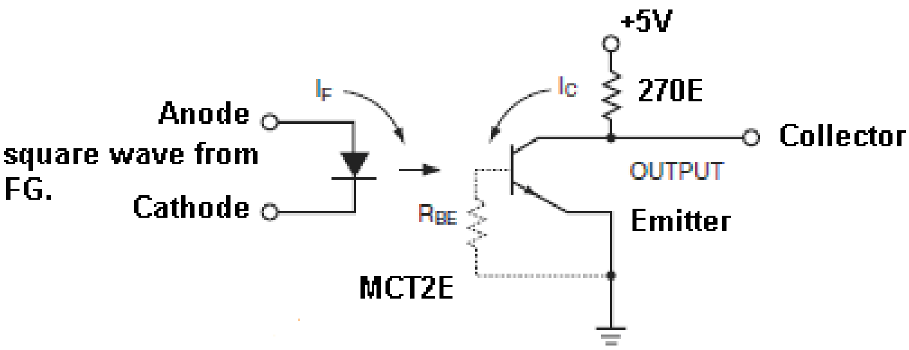
- BEL-LIT board
- Power supply
- Patch chords
- DSO
- Function generator

THEORY:-

Standard Single Channel Phototransistor Couplers. The MCT2/ MCTE family is an Industry Standard Single Channel Phototransistor. Each opto coupler consists of gallium arsenide infrared LED and a silicon NPN phototransistor.

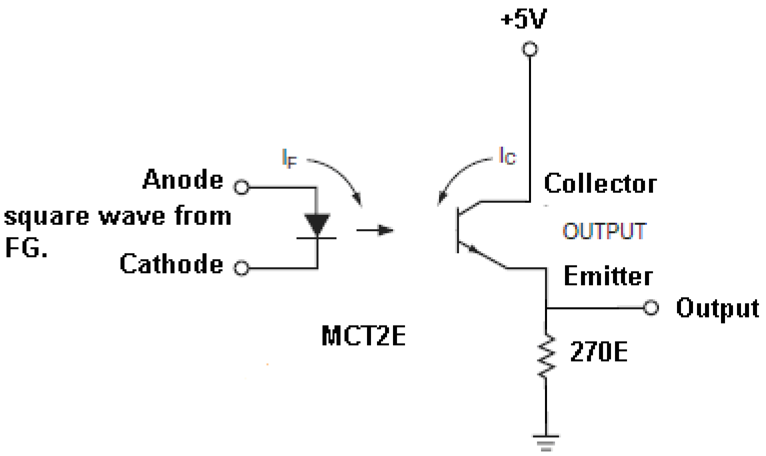
PROCEDURE:-

- Do the connection as shown in fig. below.



- Connect a square wave of 1 KHz, 3.3V dc to the anode of MCT2E. Connect cathode to the Ground point of the Function generator. Does not short board ground with Function Generator ground.
- Connect a collector resistor of 270E between supply and collector of MCT2E. Output is also taken from collector and emitter is short to board ground.
- Observe the input and output simultaneously on DSO. Observe the output by changing the input frequency.

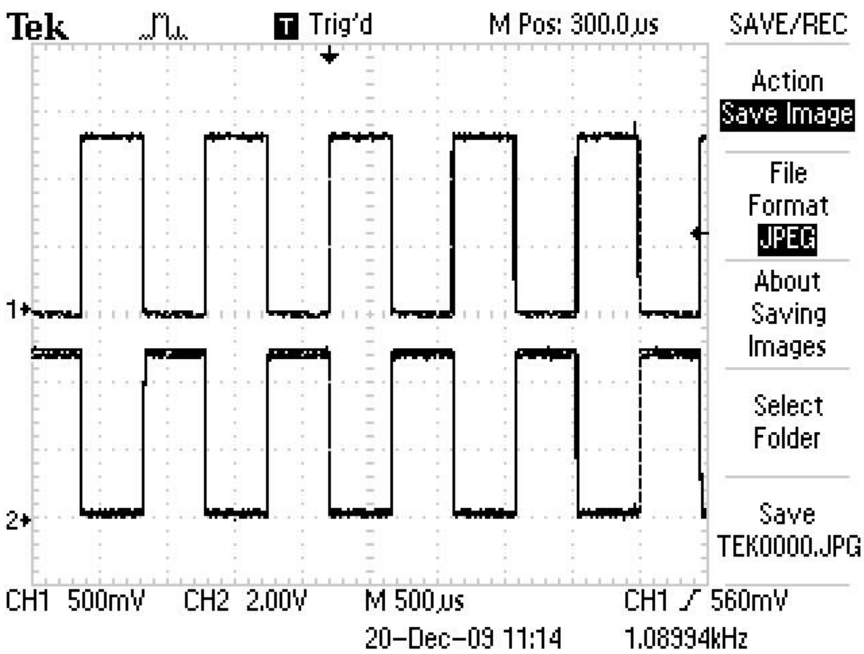
- Now change the position of resistor as shown in fig below. And observe the effect on output.
- Compare the outputs in both the configurations.



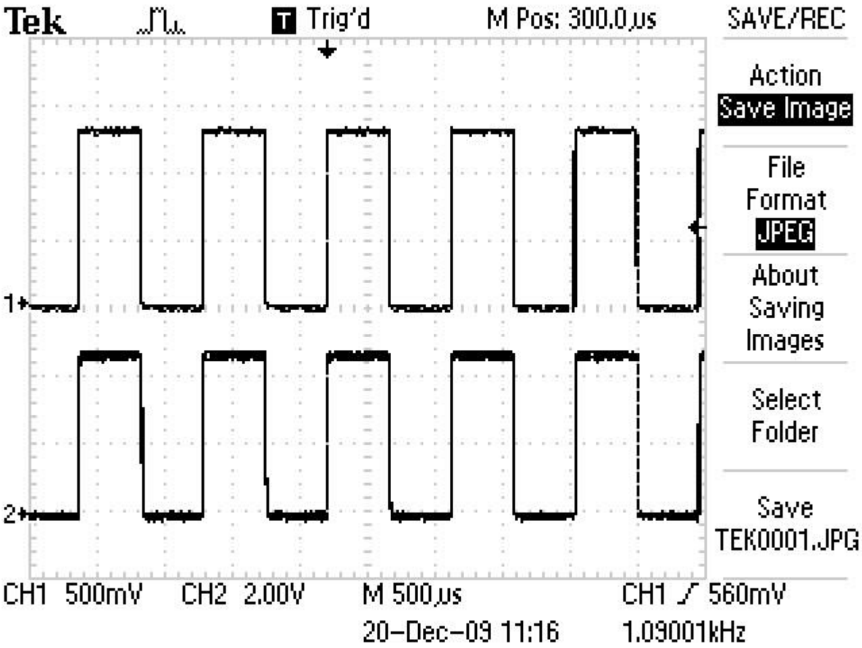
OBSERVATIONS AND WAVEFORMS:-

Using optocoupler MCT2E designer can interface two different voltage level signals from different circuits. Optocoupler is basically used for isolation purpose. It can also be used as AC mains detection, telephone ring detection, logic ground isolation, logic coupling with high frequency noise rejection etc.

- Output taken from collector



- Output taken from emitter:



EXPERIMENT NO.9

EXPERIMENT NO.9

NAME:-

To Study the Voltage Regulators LM7805 and LM7905

OBJECTIVE:-

To design and understand operation of voltage regulators LM7805 and LM7905 and study of load and line regulation

EQUIPMENTS:-

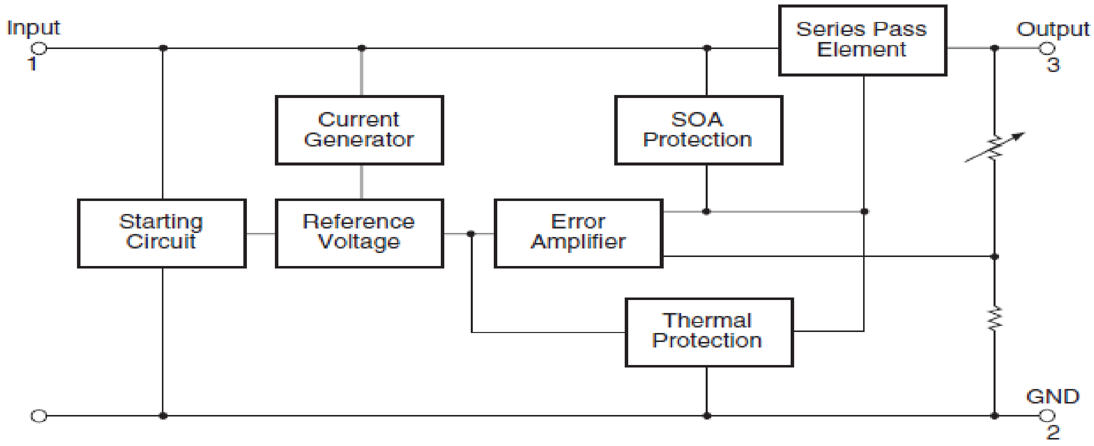
- BEL-LIT board
- Power supply
- Patch chords
- DMM (2 nos)
- Rheostat (0 to100E and 5W)
- Ammeter or DMM

THEORY:-

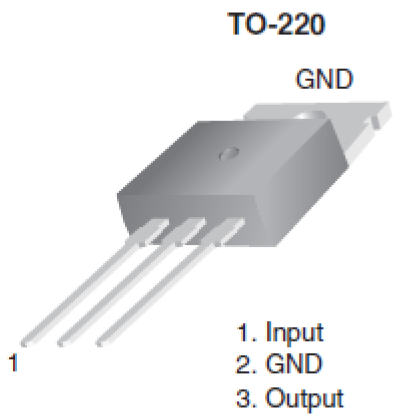
The unregulated power supply consists of transformer, rectifier and filter. The output of filter is DC voltage. If AC input to transformer fluctuates the DC voltage at the output of filter also fluctuates. The fluctuation in DC voltage may cause irregular functioning of the devices or an instrument. Unregulated power supply is the type of power supply in which the output DC voltage changes as per its input AC voltage.
In regulated power supply, the output voltage Vo remains constant & independent of 1) Change in AC input voltage. 2) Load current 3) Temperature. A voltage regulator is a circuit which supplies constant voltage under varying load and input voltage.

Positive Voltage Regulator (78XX)
The LM78XX series of three terminal positive regulators are available in the TO- 220 package and with several fixed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut down and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided; they can deliver over 1A output current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

Internal Block Diagram of 78XX



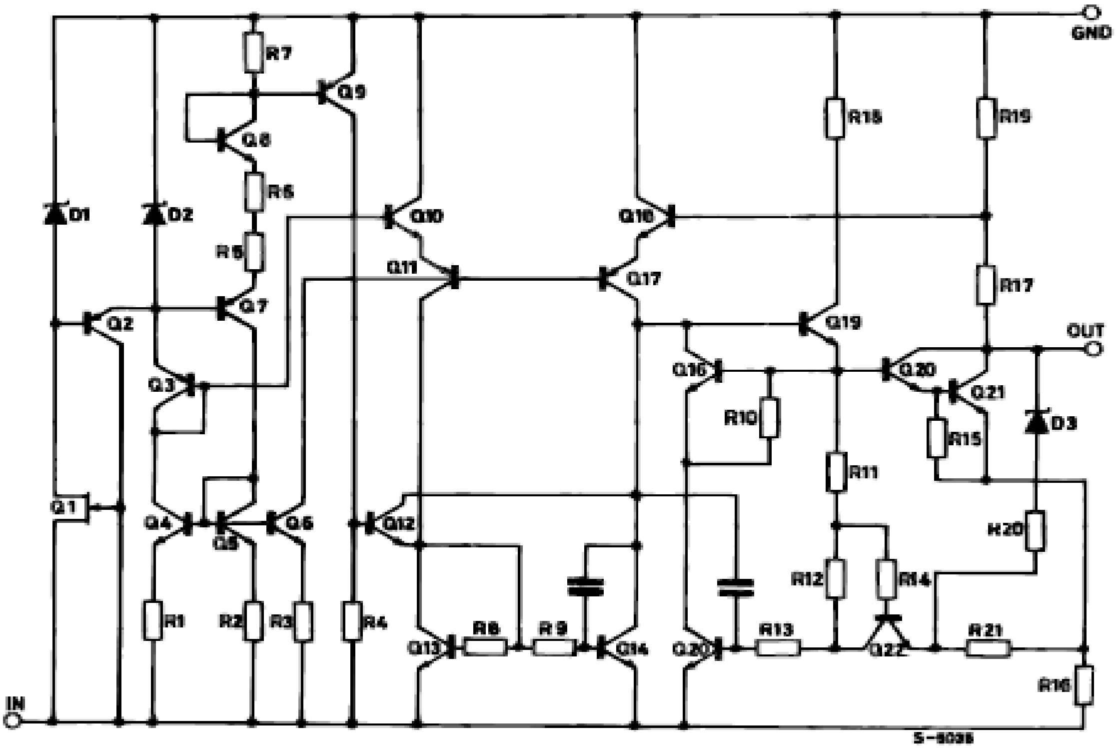
Pin Assignment of 78XX



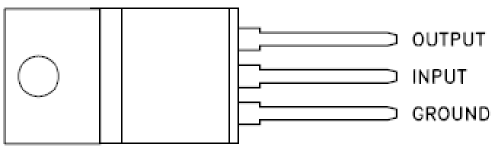
Negative Voltage Regulator (79XX)

The L7900 series of three-terminal negative regulators is available in TO-220, TO-220FP, TO-3 and D2PAK packages and several fixed output voltages, making it useful in a wide range of applications. These regulators can provide local on-card regulation, eliminating the distribution problems associated with single point regulation; furthermore, having the same voltage option as the L7800 positive standard series, they are particularly suited for split power supplies. If adequate heat sinking is provided, they can deliver over 1.5A output current.

Schematic Diagram for 79XX



Pin Assignment of 79XX



LOAD Regulation

Load regulation is the capability of regulator to maintain a constant voltage (or current) level on the output channel of a power supply despite changes in load.

Load Regulation can be defined as a percentage by the equation:

% Load Regulation = ($V_{NO\ LOAD}$ – $V_{FULL\ LOAD}$) / $V_{NO\ LOAD}$) X100% ----- Equation 1

Full Load is the load that draws the greatest current (is the lowest specified load resistance - never short circuit)

Minimum Load or No load is the load that draws the least current (is the highest specified load resistance - possibly open circuit for some types of linear supplies, usually limited by pass transistor minimum bias levels)

For switching power supplies, the primary source of regulation error is switching ripple rather than control loop inefficiency. In such cases Load Regulation is defined without normalizing to Voltage at Nominal Load and then has the units of volts.

Load Regulation, volts = Voltage (No Load) − Voltage (Full Load)

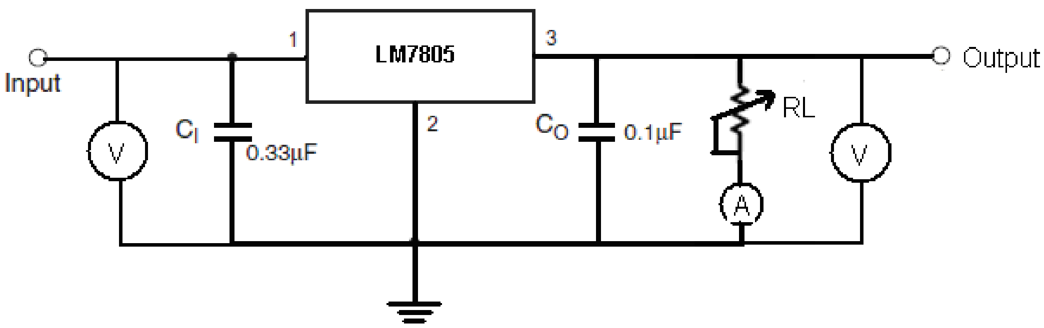
LINE Regulation

Line regulation is the capability to maintain a constant output voltage level on the output channel of a power supply despite changes to the input voltage level. Line regulation is expressed as percent of change in the output voltage relative to the change in the input line voltage

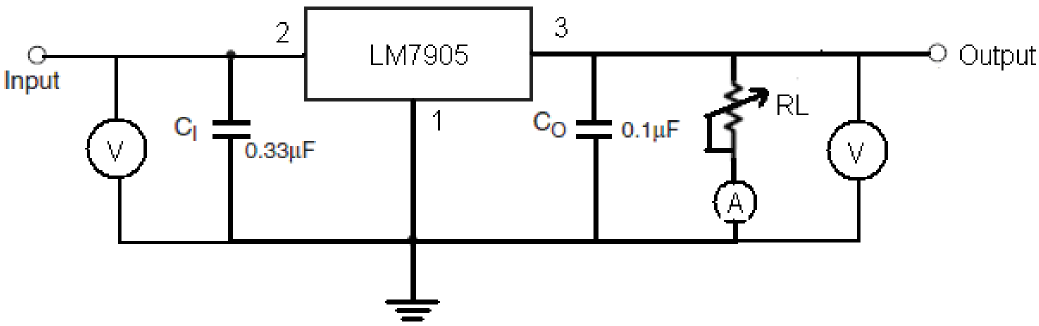
PROCEDURE:-

a) LINE Regulation

- Do the connection as shown in figure below.



- It is not required to connect decoupling capacitor because decoupling capacitors are connected from bottom side of the PCB for all regulators.
- Adjust load resistor on rheostat to 67E to draw 75mA current from the regulator under test. Connect load between output and ground as shown in fig above.
- Switch on the power supply and Adjust on board power supply 1.5V. Connect to input of the device under test (7805).
- Vary the input voltage from 1.5V to 20V and take down the corresponding output readings on DMM. Tabulate the readings.
- For line regulation of 7905 refer following figure and repeat the same procedure from 2 to 5. For 7905 apply negative voltage provided on board.
- Draw the graphs for line regulation of both 7805 and 7905.



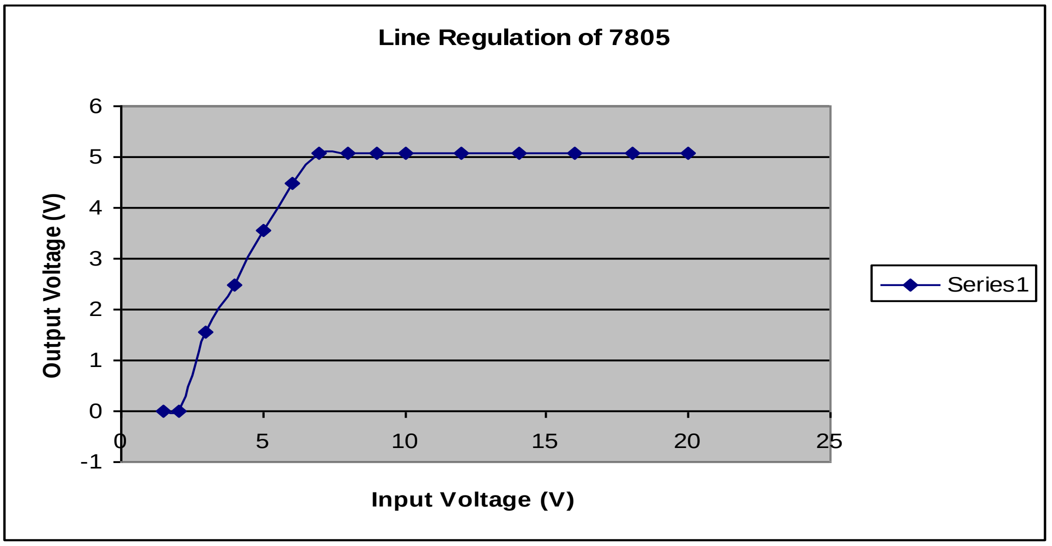
OBSERVATIONS:-

1) Line Regulation Results Observed For LM7805

SR NO	Vin (VOLTS)	Vout (VOLTS)	LOAD CURRENT (A)
1	1.5	0	0
2	2	0	0
3	3	1.54	0.024
4	4	2.47	0.036
5	5	3.54	0.053
6	6	4.5	0.066
7	7	5.08	0.075
8	8	5.08	0.075

9	9	5.08	0.075
10	10	5.08	0.075
11	12	5.08	0.075
12	14	5.08	0.075
13	16	5.08	0.075
14	18	5.08	0.075
15	20	5.08	0.075

(Table given for reference only actual values may differ from above)

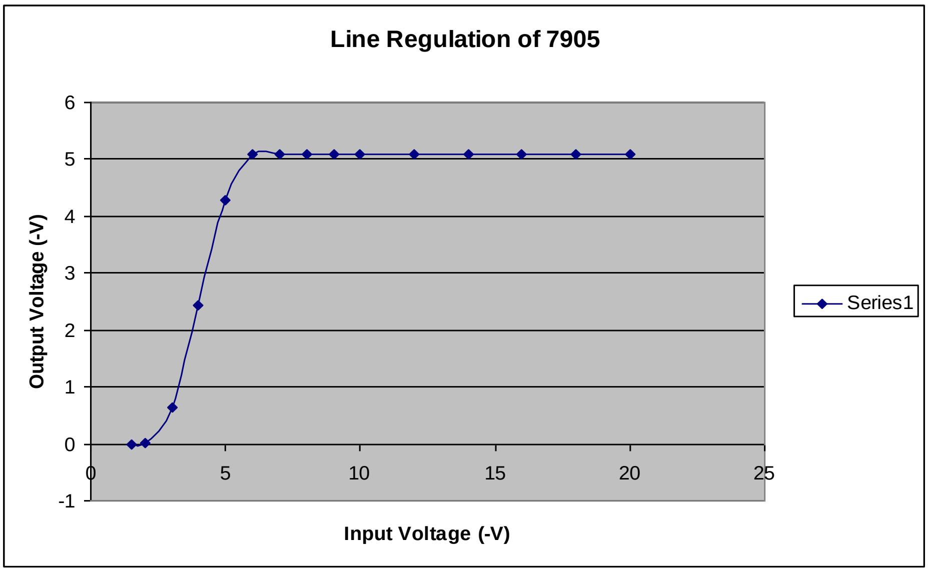


2) Line Regulation Results Observed For LM7905.

SR NO	Vin (-VOLTS)	Vout (-VOLTS)	LOAD CURRENT (A)
1	1.5	0	0
2	2	0.014	0.001
3	3	0.642	0.011
4	4	2.433	0.037
5	5	4.27	0.064
6	6	5.08	0.075
7	7	5.08	0.076
8	8	5.08	0.076
9	9	5.08	0.076
10	10	5.08	0.076
11	12	5.08	0.076
12	14	5.08	0.076
13	16	5.08	0.076
14	18	5.08	0.076
15	20	5.08	0.076

(Table given for reference only actual values may differ from above)

These are specimen readings the reading may vary board to board due to various parameter such as component tolerances, power supply of boards, temperature etc.



b) Load Regulation

Load Regulation Of LM7805

- Do the connection as per line regulation setup.
- Switch on the power supply and Adjust on board power supply 10V. Connect to input of the device under test (7805).
- Initially remove load from output and take down the reading of output voltage as $V_{NO\ LOAD}$.
- Now increase the load current to 50mA by changing the load resistance and take the reading of output voltage. Similarly take the load current VS output voltage reading for 100mA, 150mA, 200mA and 250mA.
- Calculate the load regulation of 7805 using equation 1.

Load Regulation Of LM7905

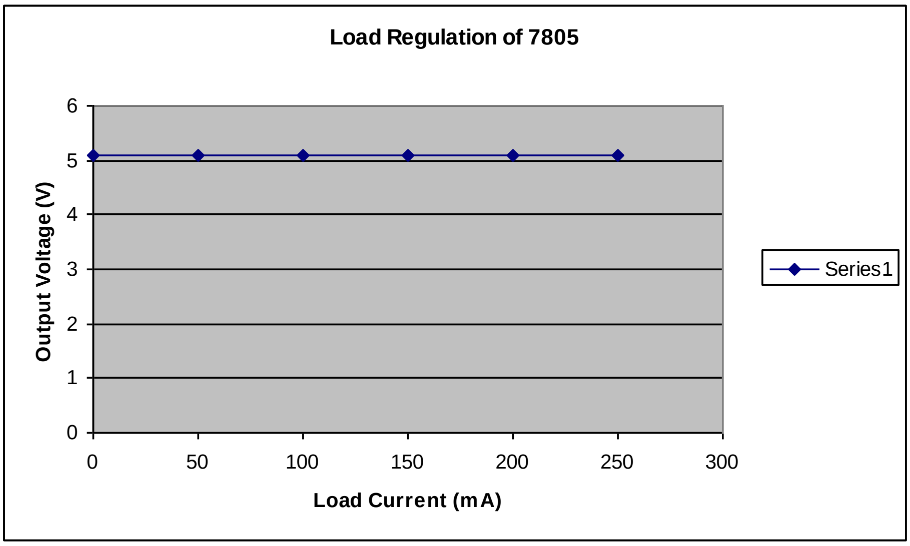
- Do the connection as per line regulation setup.
- Switch on the power supply and Adjust on board power supply -10V. Connect to input of the device under test (7905).
- Initially remove load from output and take down the reading of output voltage as $V_{NO\ LOAD}$.
- Now increase the load current to 50mA by changing the load resistance and take the reading of output voltage. Similarly take the load current VS output voltage reading for 100mA, 150mA, 200mA and 250mA.
- Calculate the load regulation of 7905 using equation 1.

OBSERVATIONS

1) Load Regulation Of LM7805

SR NO	LOAD CURRENT (mA)	OUTPUT VOLTAGE (VOLT)
1	0	5.09
2	50	5.08
3	100	5.08
4	150	5.08
5	200	5.08
6	250	5.08

(Table given for reference only actual values may differ from above)



Test conditions: input = 10v fixed. Load current is set to 250mA max.

Here our full load is at 250mA therefore for calculation consider $\mathbf{V_{Full\ load}}$ is the voltage at I_L = 250mA

$\mathbf{V_{250mA}}$ = 5.08V and

$\mathbf{V_{no\ load}}$ = 5.09V

According to equation 1 Load Regulation = ((5.09–5.08)/5.09) X 100

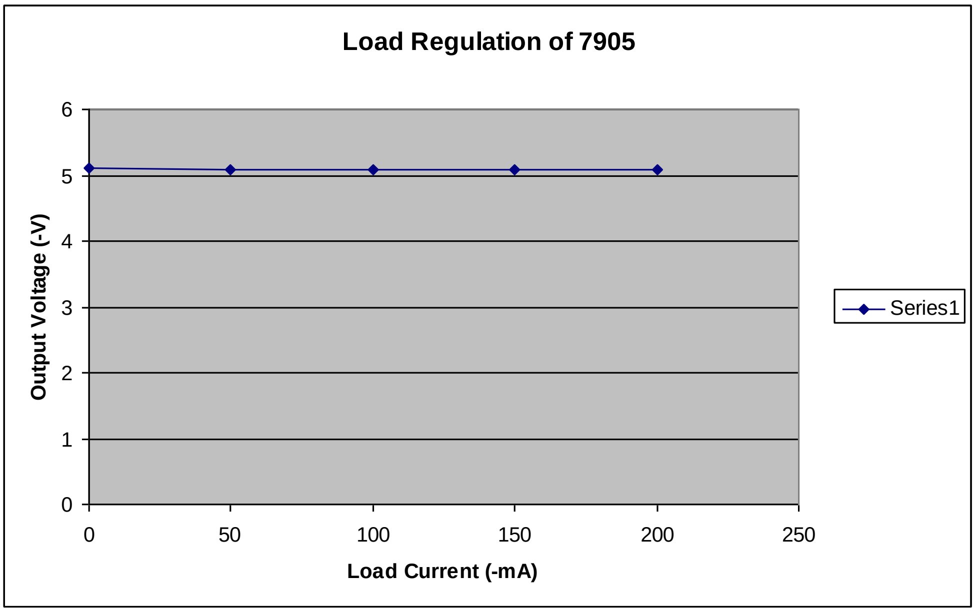
= 0.19%

Load regulation of 7805 is 0.19% for designed test conditions.

2) Load Regulation Of LM7905

SR NO	LOAD CURRENT (-mA)	OUTPUT VOLTAGE (-VOLT)
1	0	5.10
2	50	5.09
3	100	5.09
4	150	5.09
5	200	5.09

(Table given for reference only actual values may differ from above)



Test conditions: input = -10v fixed. Load current is set to -250mA max.

Here our full load is at -250mA therefore for calculation consider $\mathbf{V}_{\mathbf{Full\ load}}$ is the voltage at I_L = -250mA

$\mathbf{V}_{\mathbf{250mA}}$ = 5.09V and

$\mathbf{V}_{\mathbf{no\ load}}$ = 5.10V

According to equation 1 Load Regulation = ((5.10–5.09/5.10) X 100)

= 0.196%

Load regulation of 7905 is 0.196% for designed test conditions.

EXPERIMENT NO.10

EXPERIMENT NO.10

NAME:-

To Study the Applications of Voltage Regulators

OBJECTIVE:-

To study the voltage boost of 7805 using zener diode and resistors

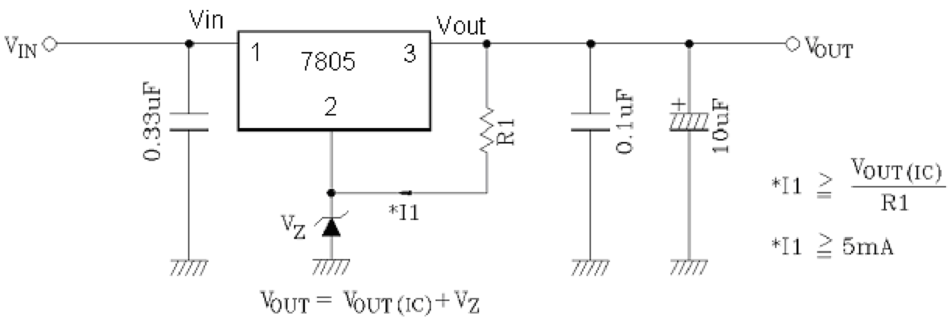
EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- DMM (02 nos)

PROCEDURE:-

A) Voltage boost of 7805 using zener diode.

- Do the connection as shown in figure below.



- Select value of R1 such that I1 is 5mA. For 7805, $Vout_{(7805)}$ = 5V. Therefore value of R1 = 5V/ 5 mA = 1K.
- Connect a zener of 2.4 V to pin 2 as show in above figure.
- According the formula the $V_{out} = Vout_{(7805)} + V_{z}$. The total output voltage = 5 + 2.4 = 7.4V.
- Switch on the power supply and apply variable voltage from 2V to 15V at input and observe the output voltage. Tabulate the readings. Compare the theoretical and practical result.

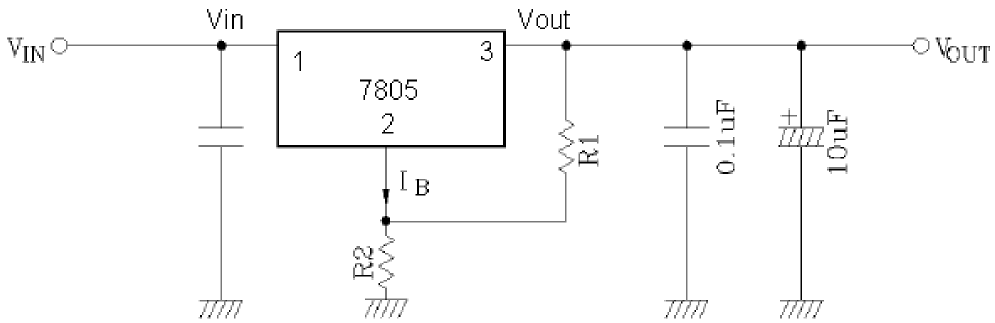
OBSERVATIONS:-

SR NO.	Vin (VOLTS)	Vout (VOLTS)
1	1.5	1.005
2	2	1.203
3	3	1.505
4	4	1.812
5	5	3.657
6	6	4.96
7	7	5.78
8	8	6.72
9	9	7.66
10	10	7.82
11	12	7.82
12	14	7.82
13	16	7.82
14	18	7.83
15	20	7.83

(Table given for reference only actual values may differ from above)

B) Voltage Boost Of 7805 Using Resistors

- Do the connection as shown in figure below.



$$V_{OUT} = V_{OUT(IC)}\left(1+\frac{R2}{R1}\right) + R2 \cdot I_B$$

-
- According to the datasheet of 7805 Quiescent current $I_{B(Typ)}$ = 4.6mA. Design output voltage to 9V.
- Assume R2 = 270E. Now calculate the value of R1 using above formula,
- 9 = 5 (1 + 270/ R1) + 270 X 4.6mA
- Thus R1 = 489E. As 489E is not standard resistor value therefore Select R1 as 470E. Taking R1= 470E Vout is around 9.1V.
- Connect variable power supply to the input of the 7805 and vary its voltage from 1.5V to 20V. Measure the output voltage of 7805 on DMM. Tabulate the readings.
- Compare the theoretical result with practical values.

OBSERVATIONS:-

SR NO.	Vin (VOLTS)	Vout (VOLTS)
1	1.5	0.014
2	2	0.020
3	3	0.915
4	4	2.70
5	5	3.76
6	6	4.65
7	7	5.66
8	8	6.67
9	9	7.62
10	10	8.65
11	12	9.01
12	14	9.02
13	16	9.03
14	18	9.04
15	20	9.05

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.11

EXPERIMENT NO.11

NAME:-

To Study the Adjustable Voltage Regulators LM317 and LM 337

OBJECTIVE:-

To design and understand operation of adjustable voltage regulators LM317 and LM337 and study of load and line regulation

EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DMM (2 nos)
- Rheostat (0 to100E and 5W)
- Ammeter or DMM.

THEORY:-

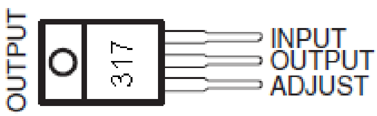
Positive Adjustable Voltage Regulator

The LM317 is an adjustable three-terminal positive-voltage regulator capable of supplying more than 1.5 A over an output-voltage range of 1.25 V to 37 V. It is exceptionally easy to use and requires only two external resistors to set the output voltage.

In addition to having higher performance than fixed regulators, this device includes on-chip current limiting, thermal overload protection, and safe operating-area protection. All overload protection remains fully functional, even if the ADJUST terminal is disconnected.

The LM317 is versatile in its applications, including uses in programmable output regulation and local on-card regulation. Or, by connecting a fixed resistor between the ADJUST and OUTPUT terminals, the LM317 can function as a precision current regulator. An optional output capacitor can be added to improve transient response. The ADJUST terminal can be bypassed to achieve very high ripple-rejection ratios, which are difficult to achieve with standard three-terminal regulators.

Pin Diagram of LM317

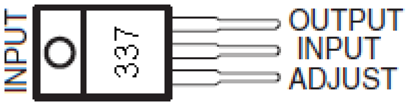


Negative Adjustable Voltage Regulator

The LM137/LM337 are adjustable 3-terminal negative voltage regulators capable of supplying in excess of −1.5A over an output voltage range of −1.2V to −37V. These regulators are exceptionally easy to apply, requiring only 2 external resistors to set the output

voltage and 1 output capacitor for frequency compensation. The circuit design has been optimized for excellent regulation and low thermal transients.
Further, the LM137 series features internal current limiting, thermal shutdown and safe-area compensation, making them virtually blowout-proof against overloads.
The LM137/LM337 serves a wide variety of applications including local on-card regulation, programmable-output voltage regulation or precision current regulation. The LM137/ LM337 are ideal complements to the LM117/LM317 adjustable positive regulators.

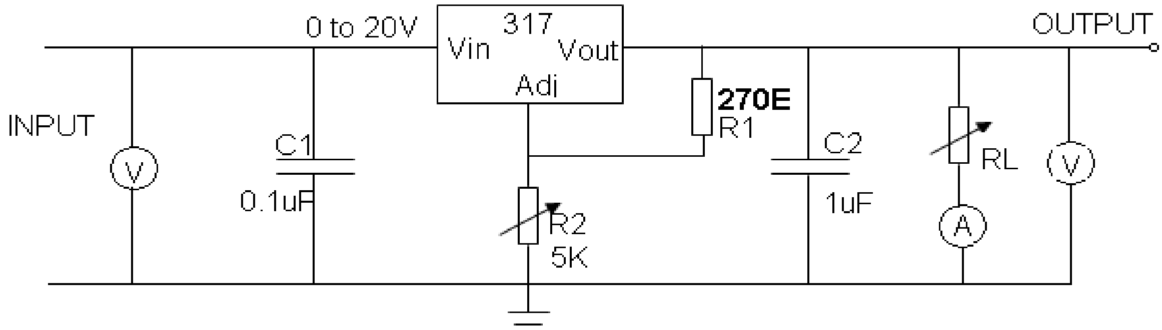
Pin Diagram of LM337



PROCEDURE:-

A) Line Regulation
I) Line Regulation Of 317

- Do the connections as shown if figure below.
- It is not required to connect decoupling capacitor because decoupling capacitors are connected from bottom side of the PCB for all regulators.



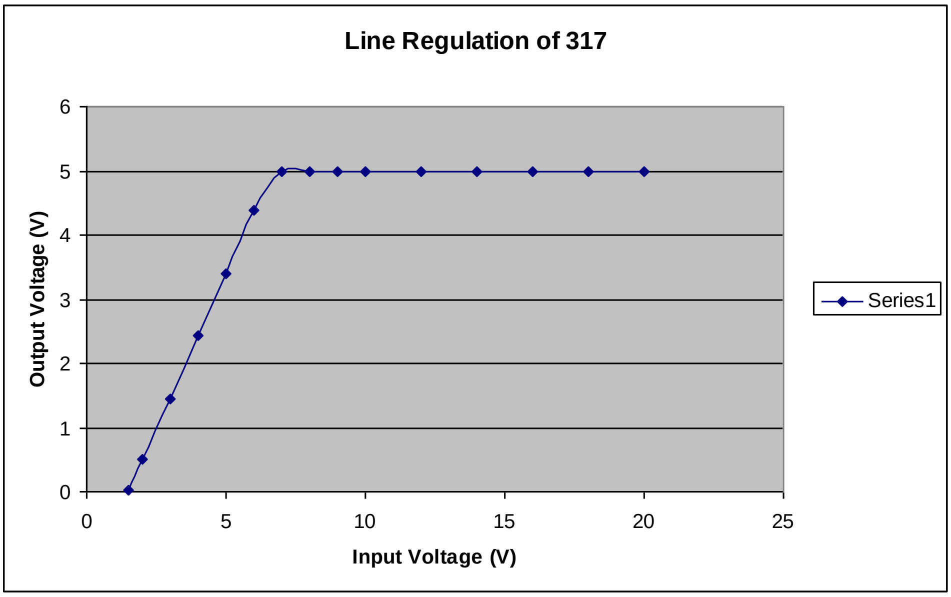
- Adjust load resistor on rheostat to 67E to draw 75mA current from the regulator under test. Connect load between output and ground as shown in fig above.
- Adjust variable power supply to 10V and connect to the input of regulator 317.
- Measure the output voltage and set it to 5V using pot at R2.
- Set variable power supply to 1.5V and increase up to 20V instep of 1 V and take down the reading of Vin VS Vout.

OBSERVATIONS:-

SR NO	Vin (VOLTS)	Vout (VOLTS)	LOAD CURRENT (A)
1	1.5	0.0232	0.003
2	2	0.501	0.006
3	3	1.456	0.022
4	4	2.428	0.036
5	5	3.40	0.051
6	6	4.38	0.065
7	7	5.00	0.075

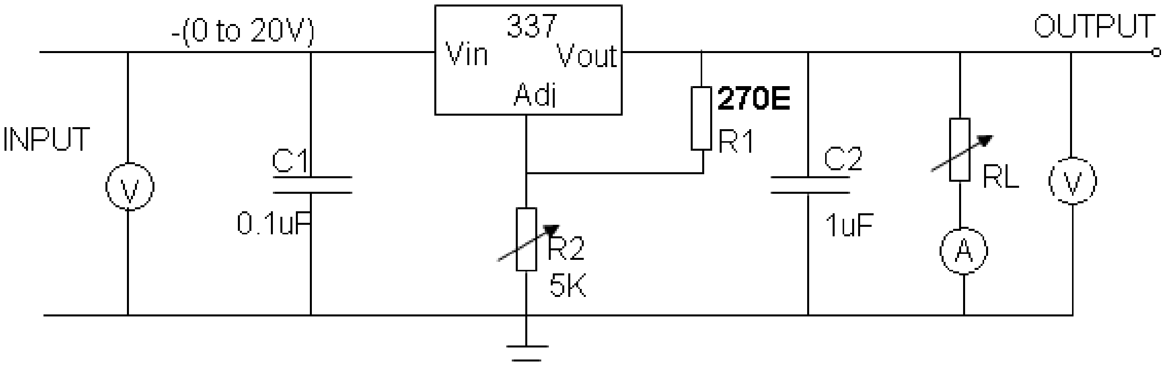
8	8	5.00	0.075
9	9	5.00	0.075
10	10	5.00	0.075
11	12	5.00	0.075
12	14	5.00	0.075
13	16	5.00	0.075
14	18	5.00	0.075
15	20	5.00	0.075

(Table given for reference only actual values may differ from above)



II) Line Regulation of 337

- Do the connections as shown if figure below.



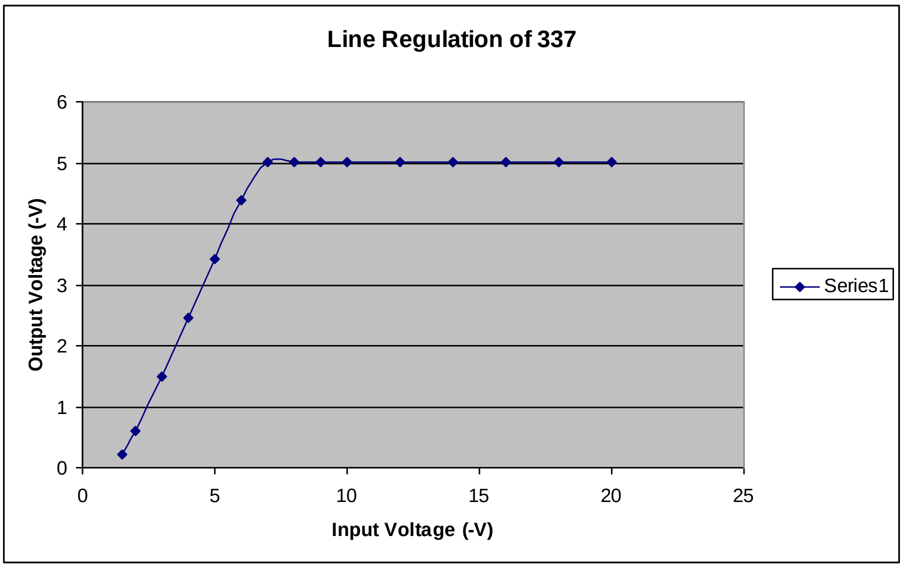
- Adjust load resistor on rheostat to 67E to draw 75mA current from the regulator under test. Connect load between output and ground as shown in fig above.
- Adjust variable power supply to -10V and connect to the input of regulator 337.
- Measure the output voltage and set it to -5V using pot at R2.
- Set variable power supply to -1.5V and increase up to -20V instep of 1 V and take down the reading of Vin VS Vout.

OBSERVATIONS:-

SR NO	Vin (-VOLTS)	Vout (-VOLTS)	LOAD CURRENT (A)
1	1.5	0.220	0.003

2	2	0.598	0.010
3	3	1.492	0.024
4	4	2.452	0.038
5	5	3.42	0.052
6	6	4.39	0.066
7	7	5.01	0.075
8	8	5.01	0.075
9	9	5.01	0.075
10	10	5.01	0.075
11	12	5.01	0.075
12	14	5.01	0.075
13	16	5.01	0.075
14	18	5.01	0.075
15	20	5.01	0.075

(Table given for reference only actual values may differ from above)



B) Load Regulation

Load Regulation Of 317

- Do the connection as per line regulation setup.
- Switch on the power supply and Adjust on board power supply 18V. Connect to input of the device under test (317) and Measure the output voltage and set it to 10V using pot at R2.
- Adjust load resistor on rheostat such that the 100 mA current is drawn from the regulator. Observe the reading of ammeter. Take down the reading of output voltage.
- Increase the load current to 150mA by changing rheostat and take down the reading of output voltage. Similarly take the output voltage readings for 200mA.
- Now remove the load and take the reading of output voltage as $V_{no\ load}$
- Calculate the load regulation of 317.

Load Regulation Of 337

- Do the connection as per line regulation setup.

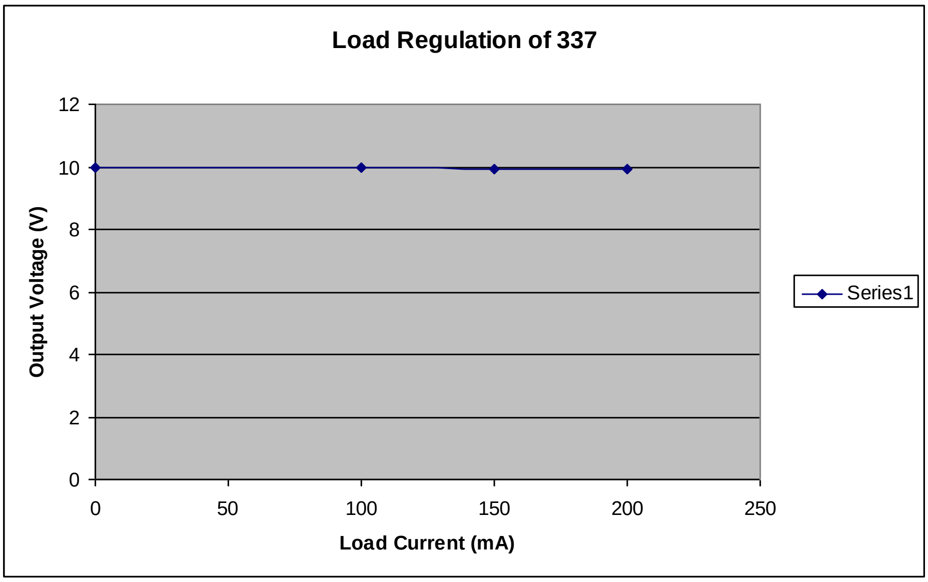
- Switch on the power supply and Adjust on board power supply -18V. Connect to input of the device under test (317) and Measure the output voltage and set it to -10V using pot at R2.
- Adjust load resistor on rheostat such that the 100mA current is drawn from the regulator. Observe the reading of ammeter. Take down the reading of output voltage. Increase the load current to 150mA by changing rheostat and take down the reading of output voltage. Similarly take the output voltage readings for 200mA.
- Now remove the load and take the reading of output voltage as $V_{no\ load}$.
- Calculate the load regulation of 337.

OBSERVATIONS:-

1) Load Regulation Of LM317

SR NO	LOAD CURRENT (mA)	OUTPUT VOLTAGE (VOLT)
1	0	9.98
2	100	9.96
3	150	9.94
4	200	9.95

(Table given for reference only actual values may differ from above)



Test conditions: input = 18V fixed. Load current is set to 200mA max.
Here our full load is at 200mA therefore for calculation consider $\mathbf{V_{Full\ load}}$ is the voltage at I_L = 200mA

$\mathbf{V_{200mA}}$ = 9.95V and
$\mathbf{V_{no\ load}}$ =9.98 V

Load Regulation = ((9.98– 9.95)/9.98) X 100
= 0.3%

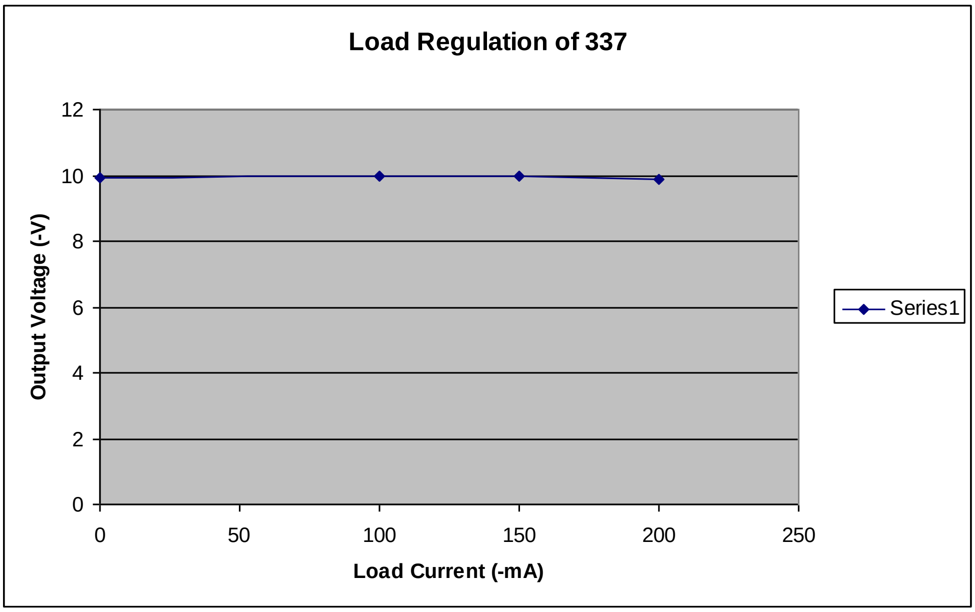
Load regulation of 317 is 0.3% for designed test conditions.

2) Load Regulation Of LM337

SR NO	LOAD CURRENT (-mA)	OUTPUT VOLTAGE (-VOLT)

1	0	9.95
2	100	9.97
3	150	9.97
4	200	9.86

(Table given for reference only actual values may differ from above)



Test conditions: input = -18V fixed. Load current is set to 200mA max.

Here our full load is at -200mA therefore for calculation consider $\mathbf{V_{Full\ load}}$ is the voltage at I_L = -200mA
$\mathbf{V_{200mA}}$ = -9.86V and
$\mathbf{V_{no\ load}}$ = -9.95V
Load Regulation = ((9.95–9.86)/9.95) X 100
= 0.9 %
Load regulation of 337 is 0.9% for designed test conditions.

EXPERIMENT NO.12

EXPERIMENT NO.12

NAME:-

To Study the Application of Adjustable Voltage Regulators 317 And 337

OBJECTIVE:-

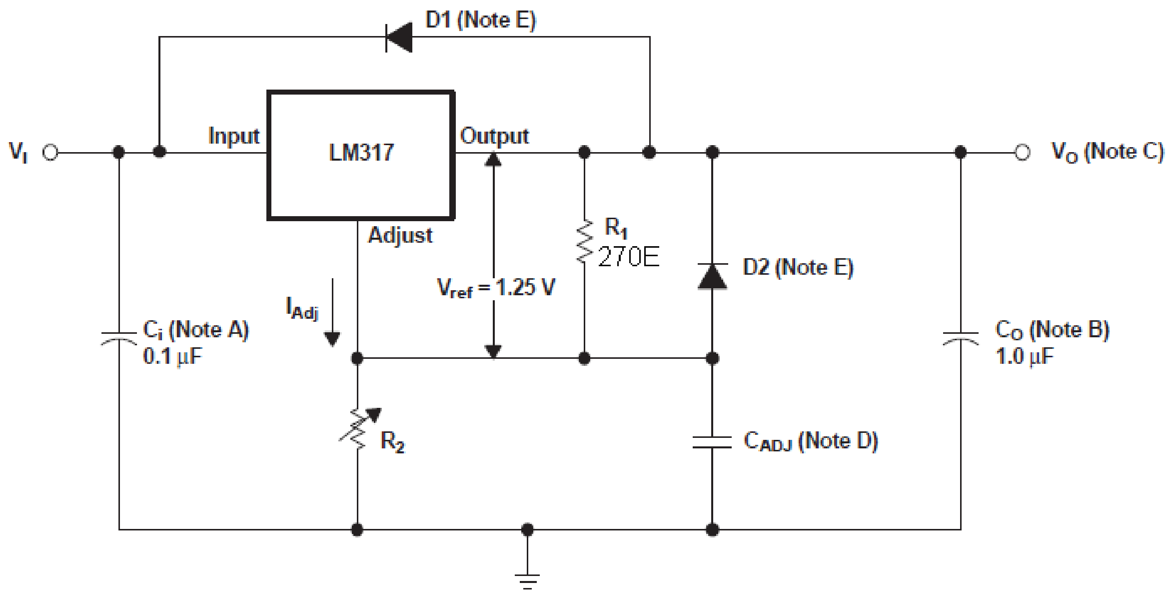
To study the operation of adjustable voltage regulator as basic variable voltage regulator with improved ripple rejection.

APPARATUS REQUIRED

- BEL-LIT board
- Power supply
- Patch chords
- DMM (2 nos)
- Rheostat (0 to100E and 5W)
- Ammeter or DMM.

THEORY

Application of 317 as variable voltage regulator with improved ripple rejection



NOTES

A. Ci is not required, but is recommended, particularly if the regulator is not in close proximity to the power-supply filter capacitors. A 0.1-uF disc or 1-uF tantalum provides sufficient bypassing for most applications, especially when adjustment and output capacitors are used.

B. C_O improves transient response, but is not needed for stability.

C. V_O is calculated as shown:

$$V_O = V_{ref}\left(1 + \frac{R_2}{R_1}\right) + (I_{Adj} \times R_2)$$

Because I_{Adj} typically is 50 mA, it is negligible in most applications.

D. C_{ADJ} is used to improve ripple rejection; it prevents amplification of the ripple as the output voltage is adjusted higher. If C_{ADJ} is used, it is best to include protection diodes.

E. If the input is shorted to ground during a fault condition, protection diodes provide measures to prevent the possibility of external capacitors discharging through low-impedance paths in the IC. By providing low-impedance discharge paths for C_O and C_{ADJ}, respectively, D1 and D2 prevent the capacitors from discharging into the output of the regulator.

PROCEDURE:-

- Do the connection as shown in figure above.
- Do not connect protection diodes and capacitors as it is connected from bottom side of the PCB.
- Connect pot of 5k at R2.
- For input and output voltage measurement connect DMM or voltmeter at input and output of device under test.
- Switch on the power supply and apply 25V from external power supply to the input of Device under test.
- Keep R2 at minimum position and measure the output voltage. Now increase the value of R2 by moving it in clockwise direction and measure the output voltage.
- Note down the readings of output voltage when R2 is at fully clockwise position.
- Verify the calculated and practical values of the output voltage.
- Do the same procedure for negative voltage regulator and note down the readings. While applying inputs and measuring outputs take care of polarities of supply.

EXPERIMENT NO.13

EXPERIMENT NO.13

NAME:-

To Study the Voltage Regulator IC 723

OBJECTIVE:-

To study the load and line regulation of voltage regulator
To study application of 723 as low and high voltage regulator.

EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DMM (2 nos), Rheostat (0 to100E and 5W)
- Ammeter or DMM.

THEORY:-

The LM723/LM723C is a voltage regulator designed primarily for series regulator applications. By itself, it will supply output currents up to 150 mA; but external transistors can be added to provide any desired load current. The circuit features extremely low standby current drain, and provision is made for either linear or fold back current limiting. The LM723/LM723C is also useful in a wide range of other applications such as a shunt regulator, a current regulator or a temperature controller

PROCEDURE:-

A) LM 723 as low voltage regulator.

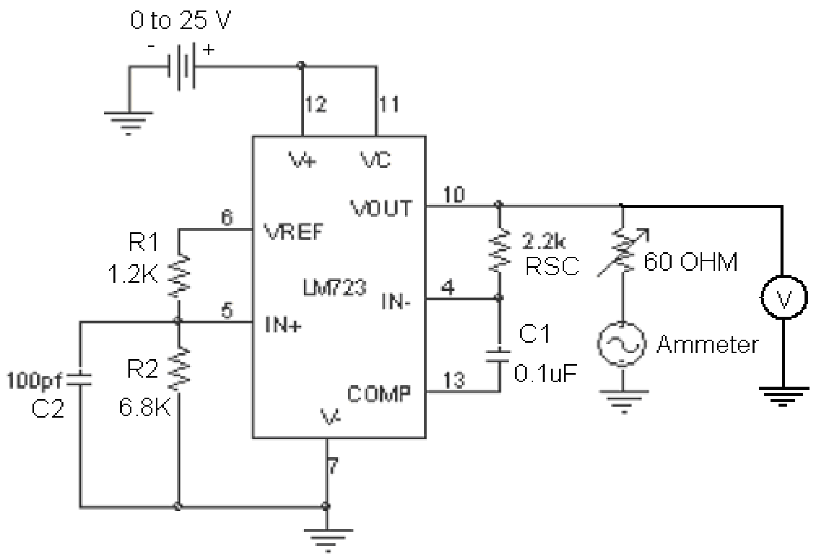


Fig. 13.1 LM723 AS LOW VOLTAGE REGULAATOR (6V)

- Do the connections as shown in fig.13.1 for designing the values refer the design procedure given below.
- Connect voltmeter and current meter at their respective places.
- Adjust rheostat to 60E before turn ON the supply so that maximum 100mA current is drawn from 723.
- Connect the supply with proper polarity and switch ON the supply.
- Vary the input voltage from 0V to 20V in step of 1V and note down the output voltage and current. This table provides the Line regulation of 723.

Design of 723 as low voltage regulator (6V)

For LM723 $V_{ref} = 7.15V$

$$V_O = 7.15\left[\frac{R_2}{R_1 + 2}\right]$$

Let the devider current I_O through the resistor R_1 & R_2 is 1mA. Since error amplifier draws very little current, we will neglect its input bias current.

$$\text{Hence } R_1 = \frac{V_{ref} - V_O}{I_D} = \frac{7.15 - 6}{1\times10^3} = 1.1K\Omega$$

$$R_2 = \frac{V_O}{I_D} = \frac{6}{1\times10^3} = 6K\Omega$$

Assume $C_1 = 0.1\mu F$ & $C_2 = 100PF$

Take R1 = 1.2K & R2 = 6.8K as standard value

- For Load regulation keep 15V input voltage to 723 and take reading of current without any load .i.e. ($V_{NO\ LOAD.}$)
- Now apply load (60E rheostat) and note down the current.i.e. ($V_{FULL\ LOAD}$)
- Calculate the Load regulation by using formula:

Load regulation = $(V_{NL} – V_{FL} / V_{NL})$ X 100%

OBSERVATIONS:-

1) Line Regulation

SR NO	Vin (VOLTS)	Vout (VOLTS)	LOAD CURRENT (A)
1	1	0.002	0.000
2	2	0.553	0.008
3	3	1.358	0.023
4	4	2.150	0.036
5	5	3.033	0.052
6	6	3.786	0.063
7	7	4.612	0.077
8	8	5.426	0.090
9	9	5.931	0.099
10	10	5.933	0.099
11	12	5.936	0.099
12	14	5.937	0.099
13	16	5.938	0.099
14	18	5.937	0.099

(Table given for reference only actual values may differ from above)

2) Load Regulation

Load regulation = $(V_{NL} – V_{FL} / V_{NL})$ X 100%
= (5.956 – 5.935 / 5.956) X 100%
= 0.35%

PROCEDURE:-

- **723 as high voltage regulator (12 V)**

DESIGN:-

Given $V_O = 12v$

$$V_O = 7.15\left[1+\frac{R_1}{R_2}\right]$$

$$12 = 7.15\left[1+\frac{R_1}{R_2}\right]$$

Assume $R_1 = 10K\Omega$

$\therefore R_2 = 17.7K\Omega$ [use $15K\Omega$]

Assume RL = 112E & C= 100pF

$$R_3 = \frac{R_1R_2}{R_1+R_2} = \underline{\quad 6K \quad}$$

Use R3 = 6.8K

- Do the connection as shown in fig. 13.2

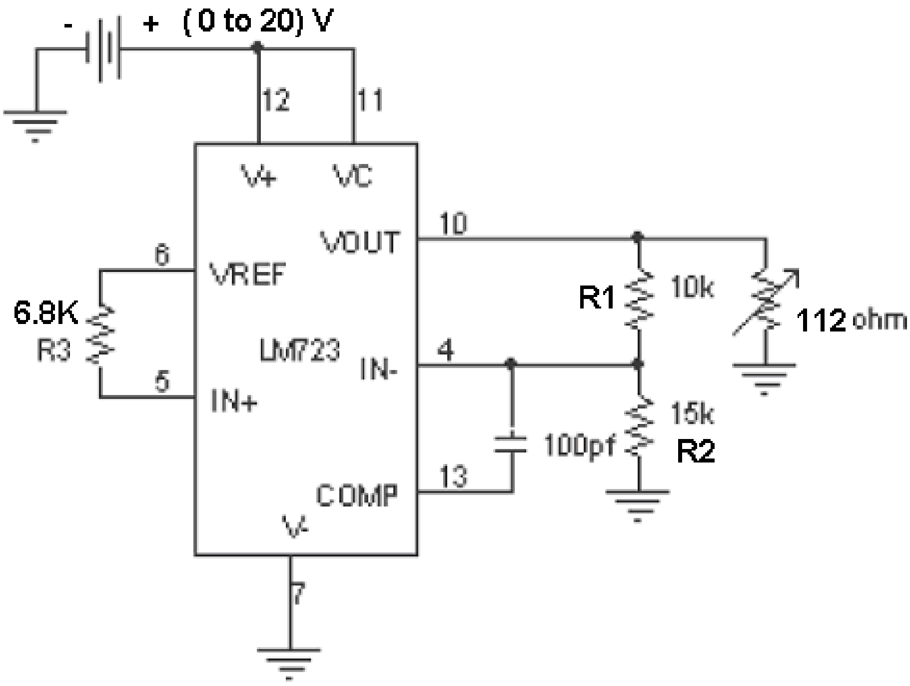


FIG- 13.2

- Connect voltmeter and current meter at their respective places.
- Adjust rheostat to 112E before turn ON the supply so that maximum 107mA current is drawn from 723.
- Connect the supply with proper polarity and switch ON the supply.
- Vary the input voltage from 0V to 20V in step of 1V and note down the output voltage and current. This table provides the Line regulation of 723.
- For Load regulation keep 20V input voltage to 723 and take reading of current without any load .i.e. ($V_{NO\ LOAD.}$)
- Now apply load (112 E rheostat) and note down the current.i.e. ($V_{FULL\ LOAD}$)
- Calculate the Load regulation by using formula:
- Load regulation = ($V_{NL} – V_{FL} / V_{NL}$) X 100%

Observations:-

1) Line Regulation

SR NO	Vin (VOLTS)	Vout (VOLTS)	LOAD CURRENT (A)
1	0	0.000	0.000
2	1	0.005	0.000
3	2	0.610	0.005
4	3	1.475	0.014
5	4	2.431	0.023
6	5	3.321	0.030
7	6	4.294	0.039
8	7	5.138	0.046
9	8	6.089	0.055
10	9	6.986	0.063
11	10	7.914	0.071
12	11	8.870	0.080

13	12	9.811	0.088
14	14	11.566	0.104
15	16	11.669	0.105
16	18	11.67	0.105
17	20	11.669	0.105

(Table given for reference only actual values may differ from above)

2) Load Regulation

Load regulation = $(V_{NL} – V_{FL} / V_{NL})$ X 100%
= (11.679 – 11.670 / 11.679) X 100%
= 0.07%

EXPERIMENT NO.14

EXPERIMENT NO. 14

NAME:-

To Study the Phase Lock Loop IC 565

OBJECTIVE:-

A) To study the design of VCO and lock and capture rage of PLL.
B) To Study PLL as a multiplier.

EQUIPMENTS:-

- BEL-LIT board
- Power supply
- Patch chords
- DSO
- Function Generator

THEORY:-

The LM565 and LM565C are general purpose phase locked loops containing a stable, highly linear voltage controlled oscillator for low distortion FM demodulation, and a double balanced phase detector with good carrier suppression.
The VCO frequency is set with an external resistor and capacitor, and a tuning range of 10:1 can be obtained with the same capacitor. The characteristics of the closed loop system bandwidth, response speed, capture and pull in range may be adjusted over a wide range with an external resistor and capacitor. The loop may be broken between the VCO and the phase detector for insertion of a digital frequency divider to obtain frequency multiplication.
In designing with phase locked loops such as the LM565, the important parameters of interest are:

Free Running Frequency

$$f_o = (0.25 / R_T C_T) \text{ Hz.} \qquad \text{-------- EQ. 1}$$

Where f_0 = VCO output frequency,
R_0 = external timing resistor,
C_0 = external timing capacitor,

A value between 2 K and 20 K is recommended for R_T. The VCO free running
Frequency is adjusted with R_T and C_T to be at the centre f_o the input frequency range.

Lock Range
The range of frequencies in the vicinity of f_o over which the VCO, once locked to the input signal, will remain locked. Lock Range is given by formula;

$f_L = \pm (8 f_o / v_{cc})$ Hz. ------- EQ. 2

Capture Range

The range of frequencies about over f_o which the loop will acquire lock with an input signal initially starting out of lock. It can be calculated by formula;

$f_C = \pm [f_L / (2\pi) (3.6) (10^3) C_2]^{½}$ -------EQ. 3

PROCEDURE:-

A) Study of VCO, Lock Range and Capture Range.

1. Connect the circuit using the component values as shown in the figure below.

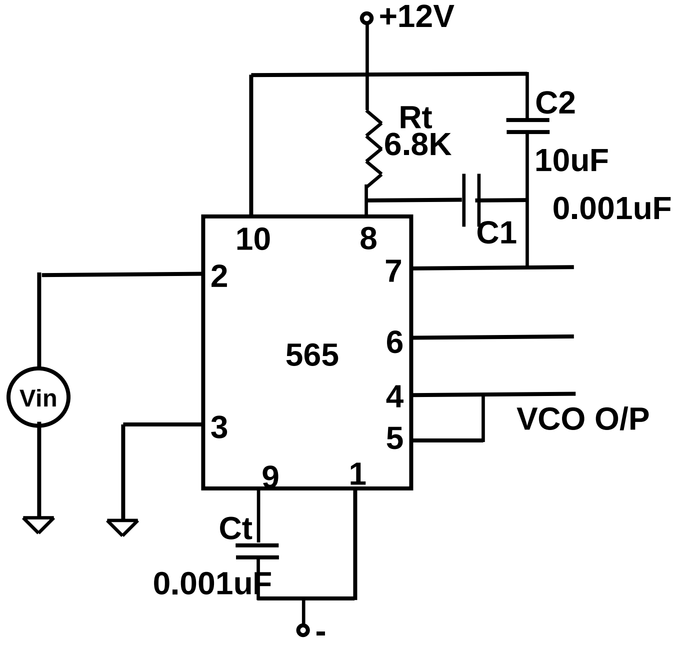


FIG. 14.1 Study of VCO, Lock range and Capture range.

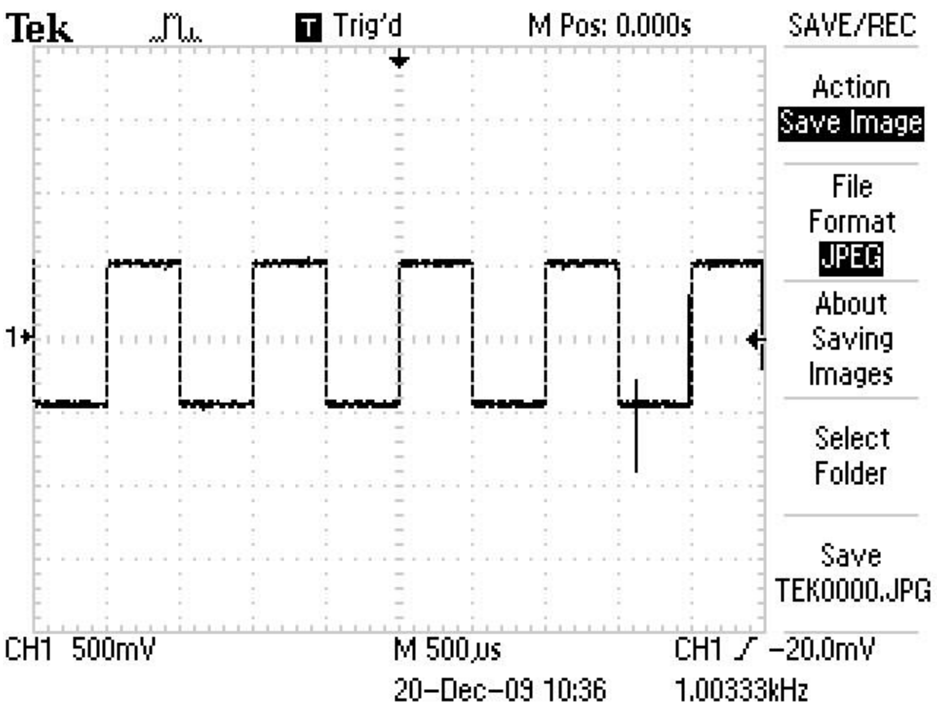
- Measure the free running frequency of VCO at pin 4 with the input signal V_{in} set to zero or ground. Compare it with the calculated value = $(0.25 / R_T C_T)$.
- Now apply the input signal of 1Vp-p square wave at 1 KHz to pin 2 of 565.
- Connect channel 1 of the scope to pin 2 and channel 2 to pin 4 of 565.
- Gradually increase the input frequency till the PLL is locked to the input frequency. This frequency f_1 gives the lower end of the capture range. Go on increase the input frequency, till PLL tracks the input signal, say to a frequency f_2. This frequency f_2 gives the upper end of the lock range. If the input frequency is increased further the loop will get unlocked.
- Now gradually decrease the input frequency till the PLL is again locked. This is the frequency f_3, the upper end of the capture range. Keep on decreasing the input frequency until the loop is unlocked. This frequency f_4 gives the lower end of the lock range.
- The lock range $\Delta f_L = (f_2 - f_4)$ compare it with the calculated value from EQ 2. Also the capture range is $\Delta f_C = (f_3 – f_1)$. Compare it with the calculated values of capture range from EQ. 3

OBSERVATIONS:-

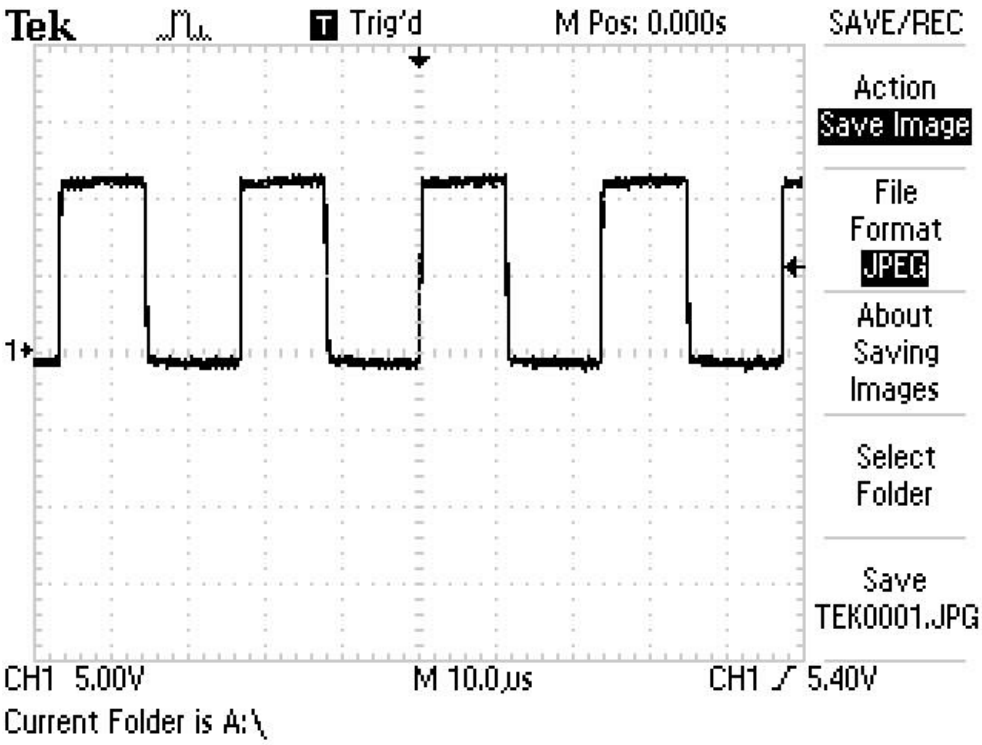
- $F_o \cong$ 37KHz. (Theoretically) and $F_o \cong$ 41.7KHz when i/p is grounded.
  $F_o \cong$ 41.3KHz when 1Vp-p square wave at 1 KHz is connected to 565.
- f_1 = 41.18 KHz. Lower end of capture range.
- f_2 = 52 KHz. Upper end of Lock Range.
- f_3 = 41.6 KHz. Upper end of capture Range.
- f_4 = 28 KHz. Lower end of Lock range.
- $\Delta f_L = (f_2 - f_4)$
  = (52 KHz – 28 KHz)
  = 24 KHz practically.
- According to EQ. 2 Δf_L = 24.66 KHz.
- $\Delta f_C = (f_3 - f_1)$
  = (41.61 KHz – 41.18 KHz)
  = 430Hz. Practically.
- According to EQ. 3 Δf_c = 330 Hz.

WAVEFORMS:-

- Input to PLL 1KHz 1Vp-p square wave.



- Centre frequency of PLL.



PROCEDURE:-

B) Study Of PLL As A Frequency Multiplier

- Connect the circuit using the component values as shown in the figure below.
- The circuit uses a 4 bit binary counter 7490 used as a divided by 5 circuit.
- Set the input signal at 1Vp-p square wave at 10KHz.
- Vary the VCO frequency by adjusting the 50K potentiometer till the PLL is locked. Measure the Phase comparator In and VCO output frequency. From these readings we can observe that PLL multiplies the input frequency by 5.

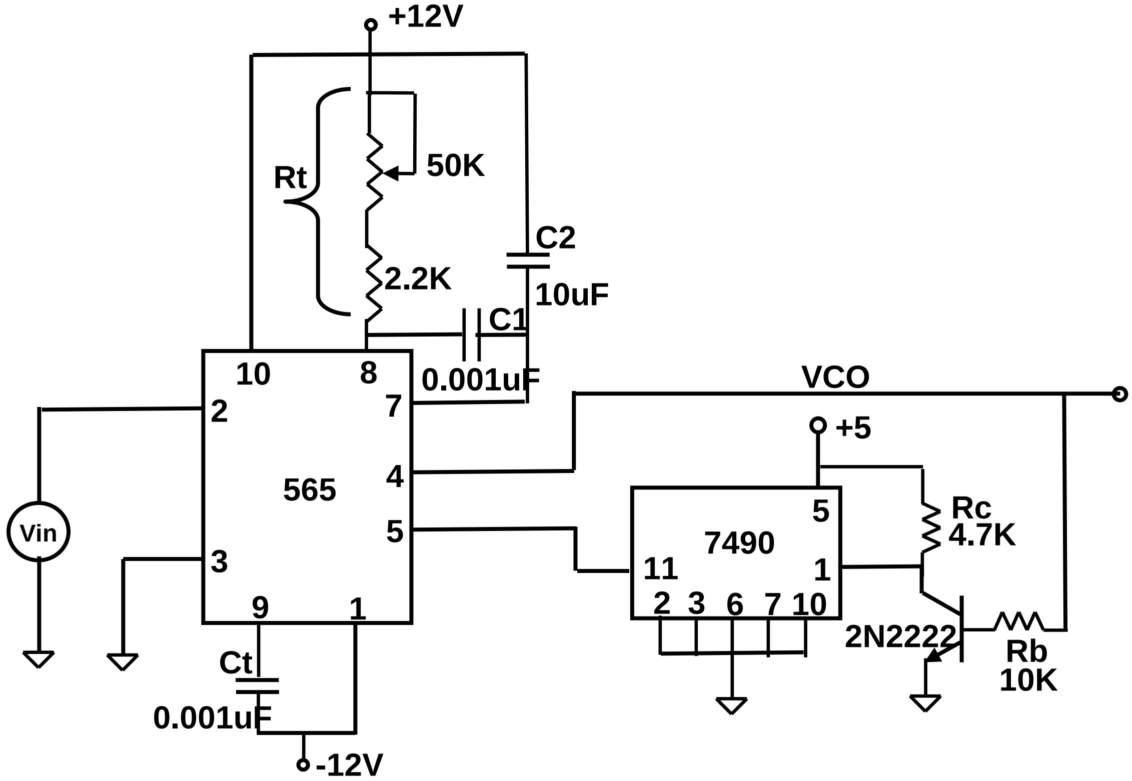
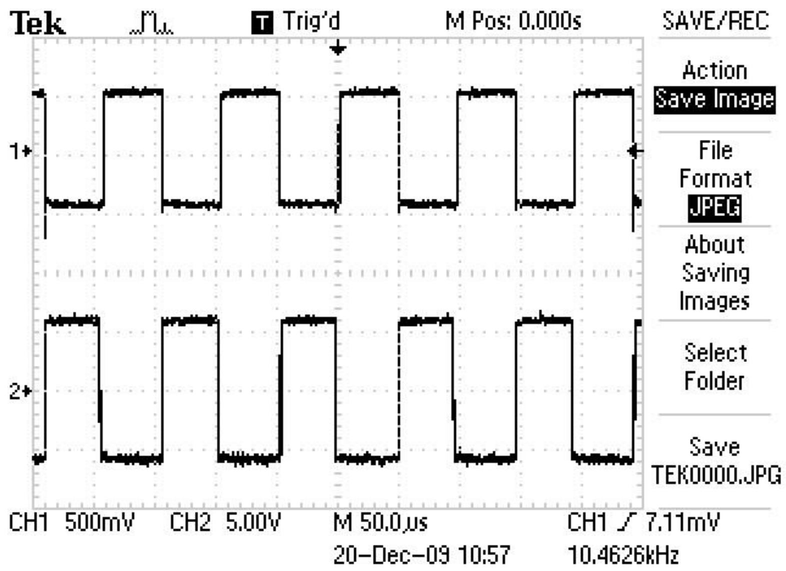


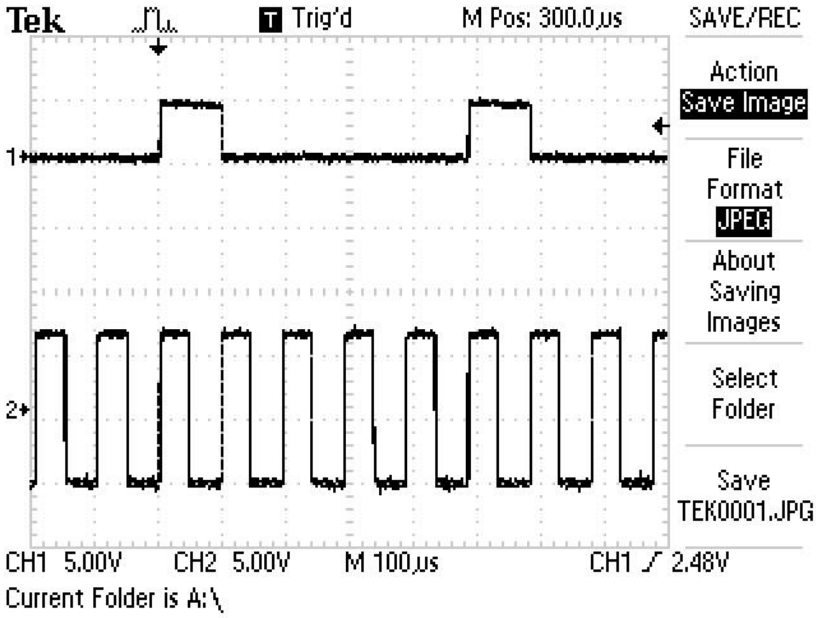
FIG. PLL as multiplier

WAVEFORMS:-

- PLL is locked to input frequency.



- PLL works as multiplier.



BEL- LIT : LINEAR IC TRAINER

RESISTOR BANK

R1	100 E	R2	100 E
R3	270 E	R4	470 E
R5	1 K	R6	1.2 K
R7	2.2 K	R8	2.7 K
R9	4.7 K	R10	5.6 K
R11	6.8 K	R12	10 K
R13	10 K	R14	12 K
R15	15 K	R16	39 K
R17	100 K	R18	200 K
R 19	220 K	R20	220 K
R21	470 K	R22	510 K

CAPACITOR BANK

C1 0.1uF DISC	C2 0.001uF DISC
C3 0.001uF DISC	C4 0.01uF DISC
C5 100pF	C6 100pF
C7 10uF ELE	C8 33pF DISC
C9 1uF ELE	C10 120pF DISC
C11 330pF DISC	C12 680pF DISC
Gc1 PF	Gc2 UF

POTENTIOMETER

POT1 1K POT2 10K

POT3 50K POT4 100K

ZENER DIODE

D1 2.4 V D2 5.1 V

TRANSISTOR BANK

Tr1 BC547 (NPN)

BASIC COMPONENT BANK